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13. Asynchronous States and the Emergence of Synchrony in Large Networks of Interacting Excitatory and Inhibitory Neurons

D. Hansel, Germán Mato

https://doi.org/10.1162/089976603321043685 | pp. 1-56

We investigate theoretically the conditions for the emergence of synchronous activity in large networks, consisting of two populations of extensively connected neurons, one excitatory and one inhibitory. The neurons are modeled with quadratic integrate-and-fire dynamics, which provide a very good approximation for the subthreshold behavior of a large class of neurons. In addition to their synaptic recurrent inputs, the neurons receive a tonic external input that varies from neuron to neuron. Because of its relative simplicity, this model can be studied analytically. We investigate the stability of the asynchronous state (AS) of the network with given average firing rates of the two populations. First, we show that the AS can remain stable even if the synaptic couplings are strong. Then we investigate the conditions under which this state can be destabilized. We show that this can happen in four generic ways. The first is a saddle-node bifurcation, which leads to another state with different average firing rates. This bifurcation, which occurs for strong enough recurrent excitation, does not correspond to the emergence of synchrony. In contrast, in the three other instability mechanisms, Hopf bifurcations, which correspond to the emergence of oscillatory synchronous activity, occur. We show that these mechanisms can be differentiated by the firing patterns they generate and their dependence on the mutual interactions of the inhibitory neurons and cross talk between the two populations. We also show that besides these codimension 1 bifurcations, the system can display several codimension 2 bifurcations: Takens-Bogdanov, Gavrielov-Guckenheimer, and double Hopf bifurcations.

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14. A Constrained EM Algorithm for Principal Component Analysis

Jong-Hoon Ahn, Jong–Hoon Oh

https://doi.org/10.1162/089976603321043694 | pp. 57-65

We propose a constrained EM algorithm for principal component analysis (PCA) using a coupled probability model derived from single-standard factor analysis models with isotropic noise structure. The single probabilistic PCA, especially for the case where there is no noise, can find only a vector set that is a linear superposition of principal components and requires postprocessing, such as diagonalization of symmetric matrices. By contrast, the proposed algorithm finds the actual principal components, which are sorted in descending order of eigenvalue size and require no additional calculation or postprocessing. The method is easily applied to kernel PCA. It is also shown that the new EM algorithm is derived from a generalized least-squares formulation.

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15. Higher-Order Statistics of Input Ensembles and the Response of Simple Model Neurons

Alexandre Kuhn, Ad Aertsen, Stefan Rotter

https://doi.org/10.1162/089976603321043702 | pp. 67-101

Pairwise correlations among spike trains recorded in vivo have been frequently reported. It has been argued that correlated activity could play an important role in the brain, because it efficiently modulates the response of a postsynaptic neuron. We show here that a neuron's output firing rate critically depends on the higher-order statistics of the input ensemble. We constructed two statistical models of populations of spiking neurons that fired with the same rates and had identical pairwise correlations, but differed with regard to the

higher-order interactions within the population. The first ensemble was characterized by clusters of spikes synchronized over the whole population. In the second ensemble, the size of spike clusters was, on average, proportional to the pairwise correlation. For both input models, we assessed the role of the size of the population, the firing rate, and the pairwise correlation on the output rate of two simple model neurons: a continuous firing-rate model and a conductance-based leaky integrate-and-fire neuron. An approximation to the mean output rate of the firing-rate neuron could be derived analytically with the help of shot noise theory. Interestingly, the essential features of the mean response of the two neuron models were similar. For both neuron models, the three input parameters played radically different roles with respect to the postsynaptic firing rate, depending on the interaction structure of the input. For instance, in the case of an ensemble with small and distributed spike clusters, the output firing rate was efficiently controlled by the size of the input population. In addition to the interaction structure, the ratio of inhibition to excitation was found to strongly modulate the effect of correlation on the postsynaptic firing rate.

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16. Duality of Rate Coding and Temporal Coding in Multilayered Feedforward Networks

Naoki Masuda, Kazuyuki Aihara

https://doi.org/10.1162/089976603321043711 | pp. 103-125

A functional role for precise spike timing has been proposed as an alternative hypothesis to rate coding. We show in this article that both the synchronous firing code and the population rate code can be used dually in a common framework of a single neural network model. Furthermore, these two coding mechanisms are bridged continuously by several modulatable model parameters, including shared connectivity, feedback strength, membrane leak rate, and neuron heterogeneity. The rates of change of these parameters are closely related to the response time and the timescale of learning.

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17. Synchronous Firing and Higher-Order Interactions in Neuron Pool

■■■-■■■ ■■■■■*, Hiroyuki Nakahara, Si Wu, Yutaka Sakai*

https://doi.org/10.1162/089976603321043720 | pp. 127-142

The stochastic mechanism of synchronous firing in a population of neurons is studied from the point of view of information geometry. Higher-order interactions of neurons, which cannot be reduced to pairwise correlations, are proved to exist in synchronous firing. In a neuron pool where each neuron fires stochastically, the probability distribution q(r) of the activity r, which is the fraction of firing neurons in the pool, is studied. When q(r) has a widespread distribution, in particular, when q(r) has two peaks, the neurons fire synchronously at one time and are quiescent at other times. The mechanism of generating such a probability distribution is interesting because the activity r is concentrated on its mean value when each neuron fires independently, because of the law of large numbers. Even when pairwise interactions, or third-order interactions, exist, the concentration is not resolved. This shows that higher-order interactions are necessary to generate widespread activity distributions. We analyze a simple model in which neurons receive common overlapping inputs and prove that such a model can have a widespread distribution of activity, generating higher-order stochastic interactions.

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18. Determination of Firing Times for the Stochastic Fitzhugh-Nagumo Neuronal Model

Henry C. Tuckwell, Roger Rodriguez, Frederic Y. M. Wan

https://doi.org/10.1162/089976603321043739 | pp. 143-159

We present for the first time an analytical approach for determining the time of firing of multicomponent nonlinear stochastic neuronal models. We apply the theory of first exit times for Markov processes to the Fitzhugh-Nagumo system with a constant mean gaussian white noise input, representing stochastic excitation and inhibition. Partial differential equations are obtained for the moments of the time to first spike. The observation that the recovery variable barely changes in the prespike trajectory leads to an accurate one-dimensional approximation. For the moments of the time to reach threshold, this leads to ordinary differential equations that may be easily solved. Several analytical approaches are explored that involve perturbation expansions for large and small values of the noise parameter. For ranges of the parameters appropriate for these asymptotic methods, the perturbation solutions are used to establish the validity of the one-dimensional approximation for both small and large values of the noise parameter. Additional verification is obtained with the excellent agreement between the mean and variance of the firing time found by numerical solution of the differential equations for the one-dimensional approximation and those obtained by simulation of the solutions of the model stochastic differential equations. Such agreement extends to intermediate values of the noise parameter. For the mean time to threshold, we find maxima at small noise values that constitute a form of stochastic resonance. We also investigate the dependence of the mean firing time on the initial values of the voltage and recovery variables when the input current has zero mean.

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19. Developmental Constraints Aid the Acquisition of Binocular Disparity Sensitivities

Melissa Dominguez, Robert A. Jacobs

https://doi.org/10.1162/089976603321043748 | pp. 161-182

This article considers the hypothesis that systems learning aspects of visual perception may benefit from the use of suitably designed developmental progressions during training. We report the results of simulations in which four models were trained to detect binocular disparities in pairs of visual images. Three of the models were developmental models in the sense that the nature of their visual input changed during the course of training. These models received a relatively impoverished visual input early in training, and the quality of this input improved as training progressed. One model used a coarse-scale-to-multiscale developmental progression, another used a fine-scale-to-multiscale progression, and the third used a random progression. The final model was nondevelopmental in the sense that the nature of its input remained the same throughout the training period. The simulation results show that the two developmental models whose progressions were organized by spatial frequency content consistently outperformed the nondevelopmental and random developmental models. We speculate that the superior performance of these two models is due to two important features of their developmental progressions: (1) these models were exposed to visual inputs at a single scale early in training, and (2) the spatial scale of their inputs progressed in an orderly fashion from one scale to a neighboring scale during training. Simulation results consistent with these speculations are presented. We conclude that suitably designed developmental sequences can be useful to systems learning to detect binocular disparities. The idea that visual development can aid visual learning is a viable hypothesis in need of study.

20. A Quantified Sensitivity Measure for Multilayer Perceptron to Input Perturbation

Xiaoqin Zeng, Daniel Yeung

https://doi.org/10.1162/089976603321043757 | pp. 183-212

The sensitivity of a neural network's output to its input perturbation is an important issue with both theoretical and practical values. In this article, we propose an approach to quantify the sensitivity of the most popular and general feedforward network: multilayer perceptron (MLP). The sensitivity measure is defined as the mathematical expectation of output deviation due to expected input deviation with respect to overall input patterns in a continuous interval. Based on the structural characteristics of the MLP, a bottom-up approach is adopted. A single neuron is considered first, and algorithms with approximately derived analytical expressions that are functions of expected input deviation are given for the computation of its sensitivity. Then another algorithm is given to compute the sensitivity of the entire MLP network. Computer simulations are used to verify the derived theoretical formulas. The agreement between theoretical and experimental results is quite good. The sensitivity measure can be used to evaluate the MLP's performance.

21. Variational Mixture of Bayesian Independent Component Analyzers

Rizwan Choudrey, Stephen Roberts

https://doi.org/10.1162/089976603321043766 | pp. 213-252

There has been growing interest in subspace data modeling over the past few years. Methods such as principal component analysis, factor analysis, and independent component analysis have gained in popularity and have found many applications in image modeling, signal processing, and data compression, to name just a few. As applications and computing power grow, more and more sophisticated analyses and meaningful representations are sought. Mixture modeling methods have been proposed for principal and factor analyzers that exploit local gaussian features in the subspace manifolds. Meaningful representations may be lost, however, if these local features are nongaussian or discontinuous. In this article, we propose extending the gaussian analyzers mixture model to an independent component analyzers mixture model. We employ recent developments in variational Bayesian inference and structure determination to construct a novel approach for modeling nongaussian, discontinuous manifolds. We automatically determine the local dimensionality of each manifold and use variational inference to calculate the optimum number of ICA components needed in our mixture model. We demonstrate our framework on complex synthetic data and illustrate its application to real data by decomposing functional magnetic resonance images into meaningful-and medically useful-features.

22. Interspike Interval Correlations, Memory, Adaptation, and Refractoriness in a Leaky Integrate-and-Fire Model with Threshold Fatigue

Maurice J. Chacron, Khashayar Pakdaman, André Longtin

https://doi.org/10.1162/089976603762552915 | pp. 253-278

Neuronal adaptation as well as interdischarge interval correlations have been shown to be functionally important properties of physiological neurons. We explore the dynamics of a modified leaky

integrate-and-fire (LIF) neuron, referred to as the LIF with threshold fatigue, and show that it reproduces these properties. In this model, the postdischarge threshold reset depends on the preceding sequence of discharge times. We show that in response to various classes of stimuli, namely, constant currents, step currents, white gaussian noise, and sinusoidal currents, the model exhibits new behavior compared with the standard LIF neuron. More precisely, (1) step currents lead to adaptation, that is, a progressive decrease of the discharge rate following the stimulus onset, while in the standard LIF, no such patterns are possible; (2) a saturation in the firing rate occurs in certain regimes, a behavior not seen in the LIF neuron; (3) interspike intervals of the noise-driven modified LIF under constant current are correlated in a way reminiscent of experimental observations, while those of the standard LIF are independent of one another; (4) the magnitude of the correlation coefficients decreases as a function of noise intensity; and (5) the dynamics of the sinusoidally forced modified LIF are described by iterates of an annulus map, an extension to the circle map dynamics displayed by the LIF model. Under certain conditions, this map can give rise to sensitivity to initial conditions and thus chaotic behavior.

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23. Reliability of Spike Timing Is a General Property of Spiking Model Neurons

Romain Brette, Emmanuel Guigon

https://doi.org/10.1162/089976603762552924 | pp. 279-308

The responses of neurons to time-varying injected currents are reproducible on a trial-by-trial basis in vitro, but when a constant current is injected, small variances in interspike intervals across trials add up, eventually leading to a high variance in spike timing. It is unclear whether this difference is due to the nature of the input currents or the intrinsic properties of the neurons. Neuron responses can fail to be reproducible in two ways: dynamical noise can accumulate over time and lead to a desynchronization over trials, or several stable responses can exist, depending on the initial condition. Here we show, through simulations and theoretical considerations, that for a general class of spiking neuron models, which includes, in particular, the leaky integrate-and-fire model as well as nonlinear spiking models, aperiodic currents, contrary to periodic currents, induce reproducible responses, which are stable under noise, change in initial conditions and deterministic perturbations of the input. We provide a theoretical explanation for aperiodic currents that cross the threshold.

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24. The Dynamic Neural Filter: A Binary Model of Spatiotemporal Coding

Brigitte Quenet, D. Horn

https://doi.org/10.1162/089976603762552933 | pp. 309-329

We describe and discuss the properties of a binary neural network that can serve as a dynamic neural filter (DNF), which maps regions of input space into spatiotemporal sequences of neuronal activity. Both deterministic and stochastic dynamics are studied, allowing the investigation of the stability of spatiotemporal sequences under noisy conditions. We define a measure of the coding capacity of a DNF and develop an algorithm for constructing a DNF that can serve as a source of given codes. On the basis of this algorithm, we suggest using a minimal DNF capable of generating observed sequences as a measure of complexity of spatiotemporal data. This measure is applied to experimental observations in the locust olfactory system, whose reverberating local field potential provides a natural temporal scale allowing the use

of a binary DNF. For random synaptic matrices, a DNF can generate very large cycles, thus becoming an efficient tool for producing spatiotemporal codes. The latter can be stabilized by applying to the parameters of the DNF a learning algorithm with suitable margins.

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25. Modeling Short-Term Synaptic Depression in Silicon

Malte Boegerhausen, Pascal Suter, Shih-Chii Liu

https://doi.org/10.1162/089976603762552942 | pp. 331-348

We describe a model of short-term synaptic depression that is derived from a circuit implementation. The dynamics of this circuit model is similar to the dynamics of some theoretical models of short-term depression except that the recovery dynamics of the variable describing the depression is nonlinear and it also depends on the presynaptic frequency. The equations describing the steady-state and transient responses of this synaptic model are compared to the experimental results obtained from a fabricated silicon network consisting of leaky integrate-and-fire neurons and different types of short-term dynamic synapses. We also show experimental data demonstrating the possible computational roles of depression. One possible role of a depressing synapse is that the input can quickly bring the neuron up to threshold when the membrane potential is close to the resting potential.

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26. Dictionary Learning Algorithms for Sparse Representation

Kenneth Kreutz-Delgado, Joseph F. Murray, Bhaskar D. Rao, Kjersti Engan, Te-Won Lee, Terrence J. Sejnowski

https://doi.org/10.1162/089976603762552951 | pp. 349-396

Algorithms for data-driven learning of domain-specific overcomplete dictionaries are developed to obtain maximum likelihood and maximum a posteriori dictionary estimates based on the use of Bayesian models with concave/Schur-concave (CSC) negative log priors. Such priors are appropriate for obtaining sparse representations of environmental signals within an appropriately chosen (environmentally matched) dictionary. The elements of the dictionary can be interpreted as concepts, features, or words capable of succinct expression of events encountered in the environment (the source of the measured signals). This is a generalization of vector quantization in that one is interested in a description involving a few dictionary entries (the proverbial "25 words or less"), but not necessarily as succinct as one entry. To learn an environmentally adapted dictionary capable of concise expression of signals generated by the environment, we develop algorithms that iterate between a representative set of sparse representations found by variants of FOCUSS and an update of the dictionary using these sparse representations. Experiments were performed using synthetic data and natural images. For complete dictionaries, we demonstrate that our algorithms have improved performance over other independent component analysis (ICA) methods, measured in terms of signal-to-noise ratios of separated sources. In the overcomplete case, we show that the true underlying dictionary and sparse sources can be accurately recovered. In tests with natural images, learned overcomplete dictionaries are shown to have higher coding efficiency than complete dictionaries; that is, images encoded with an overcomplete dictionary have both higher compression (fewer bits per pixel) and higher accuracy (lower mean square error).

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27. Spatiochromatic Receptive Field Properties Derived from Information-Theoretic Analyses of Cone Mosaic Responses to Natural Scenes

Eizaburo Doi, Toshio Inui, Te-Won Lee, Thomas Wächtler, Terrence J. Sejnowski

https://doi.org/10.1162/089976603762552960 | pp. 397-417

Neurons in the early stages of processing in the primate visual system efficiently encode natural scenes. In previous studies of the chromatic properties of natural images, the inputs were sampled on a regular array, with complete color information at every location. However, in the retina cone photoreceptors with different spectral sensitivities are arranged in a mosaic. We used an unsupervised neural network model to analyze the statistical structure of retinal cone mosaic responses to calibrated color natural images. The second-order statistical dependencies derived from the covariance matrix of the sensory signals were removed in the first stage of processing. These decorrelating filters were similar to type I receptive fields in parvo- or konio-cellular LGN in both spatial and chromatic characteristics. In the subsequent stage, the decorrelated signals were linearly transformed to make the output as statistically independent as possible, using independent component analysis. The independent component filters showed luminance selectivity with simple-cell-like receptive fields, or had strong color selectivity with large, often double-opponent, receptive fields, both of which were found in the primary visual cortex (V1). These results show that the "form" and "color" channels of the early visual system can be derived from the statistics of sensory signals.

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28. Linear Geometric ICA: Fundamentals and Algorithms

Fabian J. Theis, Andreas Jung, Carlos G. Puntonet, Elmar W. Lang

https://doi.org/10.1162/089976603762552979 | pp. 419-439

Geometric algorithms for linear independent component analysis (ICA) have recently received some attention due to their pictorial description and their relative ease of implementation. The geometric approach to ICA was proposed first by Puntonet and Prieto (1995). We will reconsider geometric ICA in a theoretic framework showing that fixed points of geometric ICA fulfill a geometric convergence condition (GCC), which the mixed images of the unit vectors satisfy too. This leads to a conjecture claiming that in the nongaussian unimodal symmetric case, there is only one stable fixed point, implying the uniqueness of geometric ICA after convergence. Guided by the principles of ordinary geometric ICA, we then present a new approach to linear geometric ICA based on histograms observing a considerable improvement in separation quality of different distributions and a sizable reduction in computational cost, by a factor of 100, compared to the ordinary geometric approach. Furthermore, we explore the accuracy of the algorithm depending on the number of samples and the choice of the mixing matrix, and compare geometric algorithms with classical ICA algorithms, namely, Extended Infomax and FastICA. Finally, we discuss the problem of high-dimensional data sets within the realm of geometrical ICA algorithms.

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29. Equivalence of Backpropagation and Contrastive Hebbian Learning in a Layered Network

Xiaohui Xie, H. Sebastian Seung

https://doi.org/10.1162/089976603762552988 | pp. 441-454

Backpropagation and contrastive Hebbian learning are two methods of training networks with hidden neurons. Backpropagation computes an error signal for the output neurons and spreads it over the hidden neurons. Contrastive Hebbian learning involves clamping the output neurons at desired values and letting the effect spread through feedback connections over the entire network. To investigate the relationship between these two forms of learning, we consider a special case in which they are identical: a multilayer perceptron with linear output units, to which weak feedback connections have been added. In this case, the change in network state caused by clamping the output neurons turns out to be the same as the error signal spread by backpropagation, except for a scalar prefactor. This suggests that the functionality of backpropagation can be realized alternatively by a Hebbian-type learning algorithm, which is suitable for implementation in biological networks.

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30. Indexed Families of Functionals and Gaussian Radial Basis Functions

Irwin W. Sandberg

https://doi.org/10.1162/089976603762552997 | pp. 455-468

We report on results concerning the capabilities of gaussian radial basis function networks in the setting of inner product spaces that need not be finite dimensional. Specifically, we show that important indexed families of functionals can be uniformly approximated, with the approximation uniform also with respect to the index. Applications are described concerning the classification of signals and the synthesis of reconfigurable classifiers.

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31. Efficient Greedy Learning of Gaussian Mixture Models

Jakob Verbeek, Nikos Vlassis, Ben Kröse

https://doi.org/10.1162/089976603762553004 | pp. 469-485

This article concerns the greedy learning of gaussian mixtures. In the greedy approach, mixture components are inserted into the mixture one after the other. We propose a heuristic for searching for the optimal component to insert. In a randomized manner, a set of candidate new components is generated. For each of these candidates, we find the locally optimal new component and insert it into the existing mixture. The resulting algorithm resolves the sensitivity to initialization of state-of-the-art methods, like expectation maximization, and has running time linear in the number of data points and quadratic in the (final) number of mixture components. Due to its greedy nature, the algorithm can be particularly useful when the optimal number of mixture components is unknown. Experimental results comparing the proposed algorithm to other methods on density estimation and texture segmentation are provided.

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32. SMO Algorithm for Least-Squares SVM Formulations

S. Sathiya Keerthi, Shirish Shevade

https://doi.org/10.1162/089976603762553013 | pp. 487-507

This article extends the well-known SMO algorithm of support vector machines (SVMs) to least-squares SVM formulations that include LS-SVM classification, kernel ridge regression, and a particular form of regularized kernel Fisher discriminant. The algorithm is shown to be asymptotically convergent. It is also

extremely easy to implement. Computational experiments show that the algorithm is fast and scales efficiently (quadratically) as a function of the number of examples.

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33. Synchronization in Networks of Excitatory and Inhibitory Neurons with Sparse, Random Connectivity

Christoph Börgers, Nancy Kopell

https://doi.org/10.1162/089976603321192059 | pp. 509-538

In model networks of E-cells and I-cells (excitatory and inhibitory neurons, respectively), synchronous rhythmic spiking often comes about from the interplay between the two cell groups: the E-cells synchronize the I-cells and vice versa. Under ideal conditions-homogeneity in relevant network parameters and all-to-all connectivity, for instance-this mechanism can yield perfect synchronization. We find that approximate, imperfect synchronization is possible even with very sparse, random connectivity. The crucial quantity is the expected number of inputs per cell. As long as it is large enough (more precisely, as long as the variance of the total number of synaptic inputs per cell is small enough), tight synchronization is possible. The desynchronizing effect of random connectivity can be reduced by strengthening the E --> I synapses. More surprising, it cannot be reduced by strengthening the I --> E synapses. However, the decay time constant of inhibition plays an important role. Faster decay yields tighter synchrony. In particular, in models in which the inhibitory synapses are assumed to be instantaneous, the effects of sparse, random connectivity cannot be seen.

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34. The Effects of Input Rate and Synchrony on a Coincidence Detector: Analytical Solution

Shawn Mikula, Ernst Niebur

https://doi.org/10.1162/089976603321192068 | pp. 539-547

We derive analytically the solution for the output rate of the ideal coincidence detector. The solution is for an arbitrary number of input spike trains with identical binomial count distributions (which includes Poisson statistics as a special case) and identical arbitrary pairwise cross-correlations, from zero correlation (independent processes) to complete correlation (identical processes).

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35. A Theoretical Model of Axon Guidance by the Robo Code

Geoffrey J. Goodhill

https://doi.org/10.1162/089976603321192077 | pp. 549-564

After crossing the midline, different populations of commissural axons in Drosophila target specific longitudinal pathways at different distances from the midline. It has recently been shown that this choice of lateral position is governed by the particular combination of Robo receptors expressed by these axons, presumably in response to a gradient of Slit released by the midline. Here we propose a simple theoretical model of this combinatorial coding scheme. The principal results of the model are that purely quantitative rather than qualitative differences between the different Robo receptors are sufficient to account for the effects observed following removal or ectopic expression of specific Robo receptors, and that the steepness

of the Slit gradient in vivo must exceed a certain minimum for the results observed experimentally to be consistent.

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36. Spike-Driven Synaptic Dynamics Generating Working Memory States

Daniel J. Amit, Gianluigi Mongillo

https://doi.org/10.1162/089976603321192086 | pp. 565-596

The collective behavior of a network, modeling a cortical module of spiking neurons connected by plastic synapses is studied. A detailed spike-driven synaptic dynamics is simulated in a large network of spiking neurons, implementing the full double dynamics of neurons and synapses. The repeated presentation of a set of external stimuli is shown to structure the network to the point of sustaining working memory (selective delay activity). When the synaptic dynamics is analyzed as a function of pre- and postsynaptic spike rates in functionally defined populations, it reveals a novel variation of the Hebbian plasticity paradigm: in any functional set of synapses between pairs of neurons (e.g., stimulated-stimulated, stimulated-delay, stimulated-spontaneous), there is a finite probability of potentiation as well as of depression. This leads to a saturation of potentiation or depression at the level of the ratio of the two probabilities. When one of the two probabilities is very high relative to the other, the familiar Hebbian mechanism is recovered. But where correlated working memory is formed, it prevents overlearning. Constraints relevant to the stability of the acquired synaptic structure and the regimes of global activity allowing for structuring are expressed in terms of the parameters describing the single-synapse dynamics. The synaptic dynamics is discussed in the light of experiments observing precise spike timing effects and related issues of biological plausibility.

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37. A Stochastic Method to Predict the Consequence of Arbitrary Forms of Spike-Timing-Dependent Plasticity

Hideyuki Câteau, Tomoki Fukai

https://doi.org/10.1162/089976603321192095 | pp. 597-620

Synapses in various neural preparations exhibit spike-timing-dependent plasticity (STDP) with a variety of learning window functions. The window functions determine the magnitude and the polarity of synaptic change according to the time difference of pre- and postsynaptic spikes. Numerical experiments revealed that STDP learning with a single-exponential window function resulted in a bimodal distribution of synaptic conductances as a consequence of competition between synapses. A slightly modified window function, however, resulted in a unimodal distribution rather than a bimodal distribution. Since various window functions have been observed in neural preparations, we develop a rigorous mathematical method to calculate the conductance distribution for any given window function. Our method is based on the Fokker-Planck equation to determine the conductance distribution and on the Ornstein-Uhlenbeck process to characterize the membrane potential fluctuations. Demonstrating that our method reproduces the known quantitative results of STDP learning, we apply the method to the type of STDP learning found recently in the CA1 region of the rat hippocampus. We find that this learning can result in nearly optimized competition between synapses. Meanwhile, we find that the type of STDP learning found in the cerebellum-like structure of electric fish can result in all-or-none synapses: either all the synaptic conductances are maximized, or none of them becomes significantly large. Our method also determines the window function that optimizes synaptic competition.

38. Permitted and Forbidden Sets in Symmetric Threshold-Linear Networks

Richard H. R. Hahnloser, H. Sebastian Seung, Jean-Jacques Slotine

https://doi.org/10.1162/089976603321192103 | pp. 621-638

The richness and complexity of recurrent cortical circuits is an inexhaustible source of inspiration for thinking about high-level biological computation. In past theoretical studies, constraints on the synaptic connection patterns of threshold-linear networks were found that guaranteed bounded network dynamics, convergence to attractive fixed points, and multistability, all fundamental aspects of cortical information processing. However, these conditions were only sufficient, and it remained unclear which were the minimal (necessary) conditions for convergence and multistability. We show that symmetric threshold-linear networks converge to a set of attractive fixed points if and only if the network matrix is copositive. Furthermore, the set of attractive fixed points is nonconnected (the network is multiattractive) if and only if the network matrix is not positive semidefinite. There are permitted sets of neurons that can be coactive at a stable steady state and forbidden sets that cannot. Permitted sets are clustered in the sense that subsets of permitted sets are permitted and supersets of forbidden sets are forbidden. By viewing permitted sets as memories stored in the synaptic connections, we provide a formulation of long-term memory that is more general than the traditional perspective of fixed-point attractor networks. There is a close correspondence between threshold-linear networks and networks defined by the generalized Lotka-Volterra equations.

39. Multistability Analysis for Recurrent Neural Networks with Unsaturating Piecewise Linear Transfer Functions

Yi Zhang, Kok Kiong Tan, T. H. Lee

https://doi.org/10.1162/089976603321192112 | pp. 639-662

Multistability is a property necessary in neural networks in order to enable certain applications (e.g., decision making), where monostable networks can be computationally restrictive. This article focuses on the analysis of multistability for a class of recurrent neural networks with unsaturating piecewise linear transfer functions. It deals fully with the three basic properties of a multistable network: boundedness, global attractivity, and complete convergence. This article makes the following contributions: conditions based on local inhibition are derived that guarantee boundedness of some multistable networks, conditions are established for global attractivity, bounds on global attractive sets are obtained, complete convergence conditions for the network are developed using novel energy-like functions, and simulation examples are employed to illustrate the theory thus developed.

40. Simple-Cell-Like Receptive Fields Maximize Temporal Coherence in Natural Video

Jarmo Hurri, Aapo Hyvärinen

https://doi.org/10.1162/089976603321192121 | pp. 663-691

Recently, statistical models of natural images have shown the emergence of several properties of the visual cortex. Most models have considered the nongaussian properties of static image patches, leading to sparse coding or independent component analysis. Here we consider the basic time dependencies of image

sequences instead of their nongaussianity. We show that simple-cell-type receptive fields emerge when temporal response strength correlation is maximized for natural image sequences. Thus, temporal response strength correlation, which is a nonlinear measure of temporal coherence, provides an alternative to sparseness in modeling simple-cell receptive field properties. Our results also suggest an interpretation of simple cells in terms of invariant coding principles, which have previously been used to explain complex-cell receptive fields.

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41. Continuous-Time Symmetric Hopfield Nets Are Computationally Universal

Ji■■■ Š■■ma, Pekka Orponen

https://doi.org/10.1162/089976603321192130 | pp. 693-733

We establish a fundamental result in the theory of computation by continuous-time dynamical systems by showing that systems corresponding to so-called continuous-time symmetric Hopfield nets are capable of general computation. As is well known, such networks have very constrained Lyapunov-function controlled dynamics. Nevertheless, we show that they are universal and efficient computational devices, in the sense that any convergent synchronous fully parallel computation by a recurrent network of n discrete-time binary neurons, with in general asymmetric coupling weights, can be simulated by a symmetric continuous-time Hopfield net containing only 18n + 7 units employing the saturated-linear activation function. Moreover, if the asymmetric network has maximum integer weight size w(max) and converges in discrete time t*, then the corresponding Hopfield net can be designed to operate in continuous time Theta(t*/epsilon) for any epsilon > 0 such that w(max)2(12n) </= epsilon2(1/epsilon). In terms of standard discrete computation models, our result implies that any polynomially space-bounded Turing machine can be simulated by a family of polynomial-size continuous-time symmetric Hopfield nets.

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42. Modeling Reverse-Phi Motion-Selective Neurons in Cortex: Double Synaptic-Veto Mechanism

Chunhui Mo, Christof Koch

https://doi.org/10.1162/08997660360581886 | pp. 735-759

Reverse-phi motion is the illusory reversal of perceived direction of movement when the stimulus contrast is reversed in successive frames. Livingstone, Tsao, and Conway (2000) showed that direction-selective cells in striate cortex of the alert macaque monkey showed reversed excitatory and inhibitory regions when two different contrast bars were flashed sequentially during a two-bar interaction analysis. While correlation or motion energy models predict the reverse-phi response, it is unclear how neurons can accomplish this. We carried out detailed biophysical simulations of a direction-selective cell model implementing a synaptic shunting scheme. Our results suggest that a simple synaptic-veto mechanism with strong direction selectivity for normal motion cannot account for the observed reverse-phi motion effect. Given the nature of reverse-phi motion, a direct interaction between the ON and OFF pathway, missing in the original shunting-inhibition model, it is essential to account for the reversal of response. We here propose a double synaptic-veto mechanism in which ON excitatory synapses are gated by both delayed ON inhibition at their null side and delayed OFF inhibition at their preferred side. The converse applies to OFF excitatory synapses. Mapping this scheme onto the dendrites of a direction-selective neuron permits the model to

respond best to normal motion in its preferred direction and to reverse-phi motion in its null direction. Two-bar interaction maps showed reversed excitation and inhibition regions when two different contrast bars are presented.

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43. Visual Development and the Acquisition of Motion Velocity Sensitivities

Robert A. Jacobs, Melissa Dominguez

https://doi.org/10.1162/08997660360581895 | pp. 761-781

We consider the hypothesis that systems learning aspects of visual perception may benefit from the use of suitably designed developmental progressions during training. Four models were trained to estimate motion velocities in sequences of visual images. Three of the models were developmental models in the sense that the nature of their visual input changed during the course of training. These models received a relatively impoverished visual input early in training, and the quality of this input improved as training progressed. One model used a coarse-to-multiscale developmental progression (it received coarse-scale motion features early in training and finer-scale features were added to its input as training progressed), another model used a fine-to-multiscale progression, and the third model used a random progression. The final model was nondevelopmental in the sense that the nature of its input remained the same throughout the training period. The simulation results show that the coarse-to-multiscale model performed best. Hypotheses are offered to account for this model's superior performance, and simulation results evaluating these hypotheses are reported. We conclude that suitably designed developmental sequences can be useful to systems learning to estimate motion velocities. The idea that visual development can aid visual learning is a viable hypothesis in need of further study.

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44. Modeling Cross-Modal Enhancement and Modality-Specific Suppression in Multisensory Neurons

Paul Patton, Thomas J. Anastasio

https://doi.org/10.1162/08997660360581903 | pp. 783-810

Cross-modal enhancement (CME) occurs when the neural response to a stimulus of one modality is augmented by another stimulus of a different modality. Paired stimuli of the same modality never produce supra-additive enhancement but may produce modality-specific suppression (MSS), in which the response to a stimulus of one modality is diminished by another stimulus of the same modality. Both CME and MSS have been described for neurons in the deep layers of the superior colliculus (DSC), but their neural mechanisms remain unknown. Previous investigators have suggested that CME involves a multiplicative amplifier, perhaps mediated by N-methyl D-aspartate (NMDA) receptors, which is engaged by cross-modal but not modality-specific input. We previously postulated that DSC neurons use multisensory input to compute the posterior probability of a target using Bayes' rule. The Bayes' rule model reproduces the major features of CME. Here we use simple neural implementations of our model to simulate both CME and MSS and to argue that multiplicative processes are not needed for CME, but may be needed to represent input variance and covariance. Producing CME requires only weighted summation of inputs and the threshold and saturation properties of simple models of biological neurons. Multiplicative nodes allow accurate computation of posterior target probabilities when the spontaneous and driven inputs have unequal variances and covariances. Neural implementations of the Bayes' rule model account better than the

multiplicative amplifier hypothesis for the effects of pharmacological blockade of NMDA receptors on the multisensory responses of DSC neurons. The neural implementations also account for MSS, given only the added hypothesis that input channels of the same modality have more spontaneous covariance than those of different modalities.

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45. Event-Driven Simulation of Spiking Neurons with Stochastic Dynamics

Jan Reutimann, Michèle Giugliano, Stefano Fusi

https://doi.org/10.1162/08997660360581912 | pp. 811-830

We present a new technique, based on a proposed event-based strategy (Mattia & Del Giudice, 2000), for efficiently simulating large networks of simple model neurons. The strategy was based on the fact that interactions among neurons occur by means of events that are well localized in time (the action potentials) and relatively rare. In the interval between two of these events, the state variables associated with a model neuron or a synapse evolved deterministically and in a predictable way. Here, we extend the event-driven simulation strategy to the case in which the dynamics of the state variables in the inter-event intervals are stochastic. This extension captures both the situation in which the simulated neurons are inherently noisy and the case in which they are embedded in a very large network and receive a huge number of random synaptic inputs. We show how to effectively include the impact of large background populations into neuronal dynamics by means of the numerical evaluation of the statistical properties of single-model neurons under random current injection. The new simulation strategy allows the study of networks of interacting neurons with an arbitrary number of external afferents and inherent stochastic dynamics.

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46. Isotropic Sequence Order Learning

Bernd Porr, Florentin Wörgötter

https://doi.org/10.1162/08997660360581921 | pp. 831-864

In this article, we present an isotropic unsupervised algorithm for temporal sequence learning. No special reward signal is used such that all inputs are completely isotropic. All input signals are bandpass filtered before converging onto a linear output neuron. All synaptic weights change according to the correlation of bandpass-filtered inputs with the derivative of the output. We investigate the algorithm in an open- and a closed-loop condition, the latter being defined by embedding the learning system into a behavioral feedback loop. In the open-loop condition, we find that the linear structure of the algorithm allows analytically calculating the shape of the weight change, which is strictly heterosynaptic and follows the shape of the weight change curves found in spike-time-dependent plasticity. Furthermore, we show that synaptic weights stabilize automatically when no more temporal differences exist between the inputs without additional normalizing measures. In the second part of this study, the algorithm is is placed in an environment that leads to closed sensor-motor loop. To this end, a robot is programmed with a prewired retraction reflex reaction in response to collisions. Through isotropic sequence order (ISO) learning, the robot achieves collision avoidance by learning the correlation between his early range-finder signals and the later occurring collision signal. Synaptic weights stabilize at the end of learning as theoretically predicted. Finally, we discuss the relation of ISO learning with other drive reinforcement models and with the commonly used temporal difference learning algorithm. This study is followed up by a mathematical analysis of the closed-loop situation in the companion article in this issue, "ISO Learning Approximates a Solution to the

Inverse-Controller Problem in an Unsupervised Behavioral Paradigm" (pp. 865-884).

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47. ISO Learning Approximates a Solution to the Inverse-Controller Problem in an Unsupervised Behavioral Paradigm

Bernd Porr, Christian von Ferber, Florentin Wörgötter

https://doi.org/10.1162/08997660360581930 | pp. 865-884

In "Isotropic Sequence Order Learning" (pp. 831-864 in this issue), we introduced a novel algorithm for temporal sequence learning (ISO learning). Here, we embed this algorithm into a formal nonevaluating (teacher free) environment, which establishes a sensor-motor feedback. The system is initially guided by a fixed reflex reaction, which has the objective disadvantage that it can react only after a disturbance has occurred. ISO learning eliminates this disadvantage by replacing the reflex-loop reactions with earlier anticipatory actions. In this article, we analytically demonstrate that this process can be understood in terms of control theory, showing that the system learns the inverse controller of its own reflex. Thereby, this system is able to learn a simple form of feedforward motor control.

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48. Localization of Function via Lesion Analysis

Ranit Aharonov, Lior Segev, Isaac Meilijson, Eytan Ruppin

https://doi.org/10.1162/08997660360581949 | pp. 885-913

This article presents a general approach for employing lesion analysis to address the fundamental challenge of localizing functions in a neural system. We describe functional contribution analysis (FCA), which assigns contribution values to the elements of the network such that the ability to predict the network's performance in response to multilesions is maximized. The approach is thoroughly examined on neurocontroller networks of evolved autonomous agents. The FCA portrays a stable set of neuronal contributions and accurate multilesion predictions that are significantly better than those obtained based on the classical single lesion approach. It is also used for a detailed synaptic analysis of the neurocontroller connectivity network, delineating its main functional backbone. The FCA provides a quantitative way of measuring how the network functions are localized and distributed among its elements. Our results question the adequacy of the classical single lesion analysis traditionally used in neuroscience and show that using lesioning experiments to decipher even simple neuronal systems requires a more rigorous multilesion analysis.

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49. The Concave-Convex Procedure

Alan Yuille, Anand Rangarajan

https://doi.org/10.1162/08997660360581958 | pp. 915-936

The concave-convex procedure (CCCP) is a way to construct discrete-time iterative dynamical systems that are guaranteed to decrease global optimization and energy functions monotonically. This procedure can be applied to almost any optimization problem, and many existing algorithms can be interpreted in terms of it. In particular, we prove that all expectation-maximization algorithms and classes of Legendre minimization and variational bounding algorithms can be reexpressed in terms of CCCP. We show that many existing neural network and mean-field theory algorithms are also examples of CCCP. The generalized iterative scaling

algorithm and Sinkhorn's algorithm can also be expressed as CCCP by changing variables. CCCP can be used both as a new way to understand, and prove the convergence of, existing optimization algorithms and as a procedure for generating new algorithms.

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50. An Analysis of Synaptic Normalization in a General Class of Hebbian Models

Terry Elliott

https://doi.org/10.1162/08997660360581967 | pp. 937-963

In standard Hebbian models of developmental synaptic plasticity, synaptic normalization must be introduced in order to constrain synaptic growth and ensure the presence of activity-dependent, competitive dynamics. In such models, multiplicative normalization cannot segregate afferents whose patterns of electrical activity are positively correlated, while subtractive normalization can. It is now widely believed that multiplicative normalization cannot segregate positively correlated afferents in any Hebbian model. However, we recently provided a counterexample to this belief by demonstrating that our own neurotrophic model of synaptic plasticity, which can segregate positively correlated afferents, can be reformulated as a nonlinear Hebbian model with competition implemented through multiplicative normalization. We now perform an analysis of a general class of Hebbian models under general forms of synaptic normalization. In particular, we extract conditions on the forms of these rules that guarantee that such models possess a fixed-point structure permitting the segregation of all but perfectly correlated afferents. We find that the failure of multiplicative normalization to segregate positively correlated afferents in a standard Hebbian model is quite atypical.

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51. Estimating a State-Space Model from Point Process Observations

Anne C. Smith, Emery N. Brown

https://doi.org/10.1162/089976603765202622 | pp. 965-991

A widely used signal processing paradigm is the state-space model. The state-space model is defined by two equations: an observation equation that describes how the hidden state or latent process is observed and a state equation that defines the evolution of the process through time. Inspired by neurophysiology experiments in which neural spiking activity is induced by an implicit (latent) stimulus, we develop an algorithm to estimate a state-space model observed through point process measurements. We represent the latent process modulating the neural spiking activity as a gaussian autoregressive model driven by an external stimulus. Given the latent process, neural spiking activity is characterized as a general point process defined by its conditional intensity function. We develop an approximate expectation-maximization (EM) algorithm to estimate the unobservable state-space process, its parameters, and the parameters of the point process. The EM algorithm combines a point process recursive nonlinear filter algorithm, the fixed interval smoothing algorithm, and the state-space covariance algorithm to compute the complete data log likelihood efficiently. We use a Kolmogorov-Smirnov test based on the time-rescaling theorem to evaluate agreement between the model and point process data. We illustrate the model with two simulated data examples: an ensemble of Poisson neurons driven by a common stimulus and a single neuron whose conditional intensity function is approximated as a local Bernoulli process.

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52. Sequential Bayesian Decoding with a Population of Neurons

Si Wu, Danmei Chen, Mahesan Niranjan, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976603765202631 | pp. 993-1012

Population coding is a simplified model of distributed information processing in the brain. This study investigates the performance and implementation of a sequential Bayesian decoding (SBD) paradigm in the framework of population coding. In the first step of decoding, when no prior knowledge is available, maximum likelihood inference is used; the result forms the prior knowledge of stimulus for the second step of decoding. Estimates are propagated sequentially to apply maximum a posteriori (MAP) decoding in which prior knowledge for any step is taken from estimates from the previous step. Not only do we analyze the performance of SBD, obtaining the optimal form of prior knowledge that achieves the best estimation result, but we also investigate its possible biological realization, in the sense that all operations are performed by the dynamics of a recurrent network. In order to achieve MAP, a crucial point is to identify a mechanism that propagates prior knowledge. We find that this could be achieved by short-term adaptation of network weights according to the Hebbian learning rule. Simulation results on both constant and time-varying stimulus support the analysis.

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53. Learning Coefficients of Layered Models When the True Distribution Mismatches the Singularities

Sumio Watanabe, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976603765202640 | pp. 1013-1033

Hierarchical learning machines such as layered neural networks have singularities in their parameter spaces. At singularities, the Fisher information matrix becomes degenerate, with the result that the conventional learning theory of regular statistical models does not hold. Recently, it was proved that if the parameter of the true distribution is contained in the singularities of the learning machine, the generalization error in Bayes estimation is asymptotically equal toλ/n, where 2λ is smaller than the dimension of the parameter andn is the number of training samples. However, the constantλ strongly depends on the local geometrical structure of singularities; hence, the generalization error is not yet clarified when the true distribution is almost but not completely contained in the singularities. In this article, in order to analyze such cases, we study the Bayes generalization error under the condition that the Kullback distance of the true distribution from the distribution represented by singularities is in proportion to 1/n and show two results. First, if the dimension of the parameter from inputs to hidden units is not larger than three, then there exists a region of true parameters such that the generalization error is larger than that of the corresponding regular model. Second, if the dimension from inputs to hidden units is larger than three, then for arbitrary true distribution, the generalization error is smaller than that of the corresponding regular model.

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54. Gamma Rhythmic Bursts: Coherence Control in Networks of Cortical Pyramidal Neurons

Toshio Aoyagi, Takashi Takekawa, Tomoki Fukai

https://doi.org/10.1162/089976603765202659 | pp. 1035-1061

Much evidence indicates that synchronized gamma-frequency (20-70 Hz) oscillation plays a significant functional role in the neocortex and hippocampus. Chattering neuron is a possible neocortical pacemaker for the gamma oscillation. Based on our recent model of chattering neurons, here we study how gamma-frequency bursting is synchronized in a network of these neurons. Using a phase oscillator description, we first examine how two coupled chattering neurons are synchronized. The analysis reveals that an incremental change of the bursting mode, such as from singlet to doublet, always accompanies a rapid transition from antisynchronous to synchronous firing. The state transition occurs regardless of what changes the bursting mode. Within each bursting mode, the neuronal activity undergoes a gradual change from synchrony to antisynchrony. Since the sensitivity to Ca(2+) and the maximum conductance of Ca(2+)-dependent cationic current as well as the intensity of input current systematically control the bursting mode, these quantities may be crucial for the regulation of the coherence of local cortical activity. Numerical simulations demonstrate that the modulations of the calcium sensitivity and the amplitude of the cationic current can induce rapid transitions between synchrony and asynchrony in a large-scale network of chattering neurons. The rapid synchronization of chattering neurons is shown to synchronize the activities of regular spiking pyramidal neurons at the gamma frequencies, as may be necessary for selective attention or binding processing in object recognition.

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55. Manhattan World: Orientation and Outlier Detection by Bayesian Inference

James M. Coughlan, Alan Yuille

https://doi.org/10.1162/089976603765202668 | pp. 1063-1088

This letter argues that many visual scenes are based on a "Manhattan" three-dimensional grid that imposes regularities on the image statistics. We construct a Bayesian model that implements this assumption and estimates the viewer orientation relative to the Manhattan grid. For many images, these estimates are good approximations to the viewer orientation (as estimated manually by the authors). These estimates also make it easy to detect outlier structures that are unaligned to the grid. To determine the applicability of the Manhattan world model, we implement a null hypothesis model that assumes that the image statistics are independent of any three-dimensional scene structure. We then use the log-likelihood ratio test to determine whether an image satisfies the Manhattan world assumption. Our results show that if an image is estimated to be Manhattan, then the Bayesian model's estimates of viewer direction are almost always accurate (according to our manual estimates), and vice versa.

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56. Kernel-Based Nonlinear Blind Source Separation

Stefan Harmeling, Andreas Ziehe, Motoaki Kawanabe, Klaus■Robert Müller

https://doi.org/10.1162/089976603765202677 | pp. 1089-1124

We propose kTDSEP, a kernel-based algorithm for nonlinear blind source separation (BSS). It combines complementary research fields: kernel feature spaces and BSS using temporal information. This yields an efficient algorithm for nonlinear BSS with invertible nonlinearity. Key assumptions are that the kernel feature space is chosen rich enough to approximate the nonlinearity and that signals of interest contain temporal information. Both assumptions are fulfilled for a wide set of real-world applications. The algorithm works as follows: First, the data are (implicitly) mapped to a high (possibly infinite)—dimensional kernel feature space.

In practice, however, the data form a smaller submanifold in feature space—even smaller than the number of training data points—a fact that has already been used by, for example, reduced set techniques for support vector machines. We propose to adapt to this effective dimension as a preprocessing step and to construct an orthonormal basis of this submanifold. The latter dimension-reduction step is essential for making the subsequent application of BSS methods computationally and numerically tractable. In the reduced space, we use a BSS algorithm that is based on second-order temporal decorrelation. Finally, we propose a selection procedure to obtain the original sources from the extracted nonlinear components automatically. Experiments demonstrate the excellent performance and efficiency of our kTDSEP algorithm for several problems of nonlinear BSS and for more than two sources.

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57. An Algorithm of Supervised Learning for Multilayer Neural Networks

Zheng Tang, Xu Gang Wang, Hiroki Tamura, Masahiro Ishii

https://doi.org/10.1162/089976603765202686 | pp. 1125-1142

A method of supervised learning for multilayer artificial neural networks to escape local minima is proposed. The learning model has two phases: a backpropagation phase and a gradient ascent phase. The backpropagation phase performs steepest descent on a surface in weight space whose height at any point in weight space is equal to an error measure, and it finds a set of weights minimizing this error measure. When the backpropagation gets stuck in local minima, the gradient ascent phase attempts to fill up the valley by modifying gain parameters in a gradient ascent direction of the error measure. The two phases are repeated until the network gets out of local minima. The algorithm has been tested on benchmark problems, such as exclusive-or (XOR), parity, alphabetic characters learning, Arabic numerals with a noise recognition problem, and a realistic real-world problem: classification of radar returns from the ionosphere. For all of these problems, the systems are shown to be capable of escaping from the backpropagation local minima and converge faster when using the new proposed method than using the simulated annealing techniques.

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58. The Time-Organized Map Algorithm: Extending the Self-Organizing Map to Spatiotemporal Signals

J. Wiemer

https://doi.org/10.1162/089976603765202695 | pp. 1143-1171

The new time-organized map (TOM) is presented for a better understanding of the self-organization and geometric structure of cortical signal representations. The algorithm extends the common self-organizing map (SOM) from the processing of purely spatial signals to the processing of spatiotemporal signals. The main additional idea of the TOM compared with the SOM is the functionally reasonable transfer of temporal signal distances into spatial signal distances in topographic neural representations. This is achieved by neural dynamics of propagating waves, allowing current and former signals to interact spatiotemporally in the neural network. Within a biologically plausible framework, the TOM algorithm (1) reveals how dynamic neural networks can self-organize to embed spatial signals in temporal context in order to realize functional meaningful invariances, (2) predicts time-organized representational structures in cortical areas representing signals with systematic temporal relation, and (3) suggests that the strength with which signals interact in the cortex determines the type of signal topology realized in topographic maps (e.g., spatially or temporally defined signal topology). Moreover, the TOM algorithm supports the explanation of topographic

reorganizations based on time-to-space transformations (Wiemer, Spengler, Joublin, Stagge, & Wacquant, 2000).

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59. New Conditions on Global Stability of Cohen-Grossberg Neural Networks

Wenlian Lu, Tianping Chen

https://doi.org/10.1162/089976603765202703 | pp. 1173-1189

In this letter, we discuss the dynamics of the Cohen-Grossberg neural networks. We provide a new and relaxed set of sufficient conditions for the Cohen-Grossberg networks to be absolutely stable and exponentially stable globally. We also provide an estimate of the rate of convergence.

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60. Estimation of Entropy and Mutual Information

Liam Paninski

https://doi.org/10.1162/089976603321780272 | pp. 1191-1253

We present some new results on the nonparametric estimation of entropy and mutual information. First, we use an exact local expansion of the entropy function to prove almost sure consistency and central limit theorems for three of the most commonly used discretized information estimators. The setup is related to Grenander's method of sieves and places no assumptions on the underlying probability measure generating the data. Second, we prove a converse to these consistency theorems, demonstrating that a misapplication of the most common estimation techniques leads to an arbitrarily poor estimate of the true information, even given unlimited data. This “inconsistency” theorem leads to an analytical approximation of the bias, valid in surprisingly small sample regimes and more accurate than the usual [Formula: see text] formula of Miller and Madow over a large region of parameter space. The two most practical implications of these results are negative: (1) information estimates in a certain data regime are likely contaminated by bias, even if “bias-corrected” estimators are used, and (2) confidence intervals calculated by standard techniques drastically underestimate the error of the most common estimation methods. Finally, we note a very useful connection between the bias of entropy estimators and a certain polynomial approximation problem. By casting bias calculation problems in this approximation theory framework, we obtain the best possible generalization of known asymptotic bias results. More interesting, this framework leads to an estimator with some nice properties: the estimator comes equipped with rigorous bounds on the maximum error over all possible underlying probability distributions, and this maximum error turns out to be surprisingly small. We demonstrate the application of this new estimator on both real and simulated data.

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61. A Taxonomy for Spatiotemporal Connectionist Networks Revisited: The Unsupervised Case

Guilherme A. Barreto, A.F.R. Araujo, Stefan C. Kremer

https://doi.org/10.1162/089976603321780281 | pp. 1255-1320

Spatiotemporal connectionist networks (STCNs) comprise an important class of neural models that can deal with patterns distributed in both time and space. In this article, we widen the application domain of the taxonomy for supervised STCNs recently proposed by Kremer (2001) to the unsupervised case. This is

possible through a reinterpretation of the state vector as a vector of latent (hidden) variables, as proposed by Meinicke (2000). The goal of this generalized taxonomy is then to provide a nonlinear generative framework for describing unsupervised spatiotemporal networks, making it easier to compare and contrast their representational and operational characteristics. Computational properties, representational issues, and learning are also discussed, and a number of references to the relevant source publications are provided. It is argued that the proposed approach is simple and more powerful than the previous attempts from a descriptive and predictive viewpoint. We also discuss the relation of this taxonomy with automata theory and state-space modeling and suggest directions for further work.

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62. On Embedding Synfire Chains in a Balanced Network

Yuval Aviel, Carsten Mehring, Micha Abeles, D. Horn

https://doi.org/10.1162/089976603321780290 | pp. 1321-1340

We investigate the formation of synfire waves in a balanced network of integrate-and-fire neurons. The synaptic connectivity of this network embodies synfire chains within a sparse random connectivity. This network can exhibit global oscillations but can also operate in an asynchronous activity mode. We analyze the correlations of two neurons in a pool as convenient indicators for the state of the network. We find, using different models, that these indicators depend on a scaling variable. Beyond a critical point, strong correlations and large network oscillations are obtained. We looked for the conditions under which a synfire wave could be propagated on top of an otherwise asynchronous state of the network. This condition was found to be highly restrictive, requiring a large number of neurons for its implementation in our network. The results are based on analytic derivations and simulations.

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63. Ergodicity of Spike Trains: When Does Trial Averaging Make Sense?

Naoki Masuda, Kazuyuki Aihara

https://doi.org/10.1162/089976603321780308 | pp. 1341-1372

Neuronal information processing is often studied on the basis of spiking patterns. The relevant statistics such as firing rates calculated with the peri-stimulus time histogram are obtained by averaging spiking patterns over many experimental runs. However, animals should respond to one experimental stimulation in real situations, and what is available to the brain is not the trial statistics but the population statistics. Consequently, physiological ergodicity, namely, the consistency between trial averaging and population averaging, is implicitly assumed in the data analyses, although it does not trivially hold true. In this letter, we investigate how characteristics of noisy neural network models, such as single neuron properties, external stimuli, and synaptic inputs, affect the statistics of firing patterns. In particular, we show that how high membrane potential sensitivity to input fluctuations, inability of neurons to remember past inputs, external stimuli with large variability and temporally separated peaks, and relatively few contributions of synaptic inputs result in spike trains that are reproducible over many trials. The reproducibility of spike trains and synchronous firing are contrasted and related to the ergodicity issue. Several numerical calculations with neural network examples are carried out to support the theoretical results.

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64. Laplacian Eigenmaps for Dimensionality Reduction and Data Representation

Mikhail Belkin, Partha Niyogi

https://doi.org/10.1162/089976603321780317 | pp. 1373-1396

One of the central problems in machine learning and pattern recognition is to develop appropriate representations for complex data. We consider the problem of constructing a representation for data lying on a low-dimensional manifold embedded in a high-dimensional space. Drawing on the correspondence between the graph Laplacian, the Laplace Beltrami operator on the manifold, and the connections to the heat equation, we propose a geometrically motivated algorithm for representing the high-dimensional data. The algorithm provides a computationally efficient approach to nonlinear dimensionality reduction that has locality-preserving properties and a natural connection to clustering. Some potential applications and illustrative examples are discussed.

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65. Leave-One-Out Bounds for Kernel Methods

Tong Zhang

https://doi.org/10.1162/089976603321780326 | pp. 1397-1437

In this article, we study leave-one-out style cross-validation bounds for kernel methods. The essential element in our analysis is a bound on the parameter estimation stability for regularized kernel formulations. Using this result, we derive bounds on expected leave-one-out cross-validation errors, which lead to expected generalization bounds for various kernel algorithms. In addition, we also obtain variance bounds for leave-oneout errors. We apply our analysis to some classification and regression problems and compare them with previous results.

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66. Background Synaptic Activity as a Switch Between Dynamical States in a Network

Emilio Salinas

https://doi.org/10.1162/089976603321891756 | pp. 1439-1475

A bright red light may trigger a sudden motor action in a driver crossing an intersection: stepping at once on the brakes. The same red light, however, may be entirely inconsequential if it appears, say, inside a movie theater. Clearly, context determines whether a particular stimulus will trigger a motor response, but what is the neural correlate of this? How does the nervous system enable or disable whole networks so that they are responsive or not to a given sensory signal? Using theoretical models and computer simulations, I show that networks of neurons have a built-in capacity to switch between two types of dynamic state: one in which activity is low and approximately equal for all units, and another in which different activity distributions are possible and may even change dynamically. This property allows whole circuits to be turned on or off by weak, unstructured inputs. These results are illustrated using networks of integrate-and-fire neurons with diverse architectures. In agreement with the analytic calculations, a uniform background input may determine whether a random network has one or two stable firing levels; it may give rise to randomly alternating firing episodes in a circuit with reciprocal inhibition; and it may regulate the capacity of a center-surround circuit to produce either self-sustained activity or traveling waves. Thus, the functional

properties of a network may be drastically modified by a simple, weak signal. This mechanism works as long as the network is able to exhibit stable firing states, or attractors.

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67. Note on “Comparison of Model Selection for Regression” by Vladimir Cherkassky and Yunqian Ma

Trevor Hastie, Rob Tibshirani, Jerome H. Friedman

https://doi.org/10.1162/089976603321891765 | pp. 1477-1480

While Cherkassky and Ma (2003) raise some interesting issues in comparing techniques for model selection, their article appears to be written largely in protest of comparisons made in our book, Elements of Statistical Learning (2001). Cherkassky and Ma feel that we falsely represented the structural risk minimization (SRM) method, which they defend strongly here. In a two-page section of our book (pp. 212-213), we made an honest attempt to compare the SRM method with two related techniques, Aikaike information criterion (AIC) and Bayesian information criterion (BIC). Apparently, we did not apply SRM in the optimal way. We are also accused of using contrived examples, designed to make SRM look bad. Alas, we did introduce some careless errors in our original simulation--errors that were corrected in the second and subsequent printings. Some of these errors were pointed out to us by Cherkassky and Ma (we supplied them with our source code), and as a result we replaced the assessment "SRM performs poorly overall" with a more moderate "the performance of SRM is mixed" (p. 212).

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68. Spike-Timing-Dependent Plasticity and Relevant Mutual Information Maximization

Gal Chechik

https://doi.org/10.1162/089976603321891774 | pp. 1481-1510

Synaptic plasticity was recently shown to depend on the relative timing of the pre- and postsynaptic spikes. This article analytically derives a spike-dependent learning rule based on the principle of information maximization for a single neuron with spiking inputs. This rule is then transformed into a biologically feasible rule, which is compared to the experimentally observed plasticity. This comparison reveals that the biological rule increases information to a near-optimal level and provides insights into the structure of biological plasticity. It shows that the time dependency of synaptic potentiation should be determined by the synaptic transfer function and membrane leak. Potentiation consists of weight-dependent and weight-independent components whose weights are of the same order of magnitude. It further suggests that synaptic depression should be triggered by rare and relevant inputs but at the same time serves to unlearn the baseline statistics of the network's inputs. The optimal depression curve is uniformly extended in time, but biological constraints that cause the cell to forget past events may lead to a different shape, which is not specified by our current model. The structure of the optimal rule thus suggests a computational account for several temporal characteristics of the biological spike-timing-dependent rules.

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69. Relating STDP to BCM

Eugene M. Izhikevich, Niraj S. Desai

https://doi.org/10.1162/089976603321891783 | pp. 1511-1523

We demonstrate that the BCM learning rule follows directly from STDP when pre- and postsynaptic neurons fire uncorrelated or weakly correlated Poisson spike trains, and only nearest-neighbor spike interactions are taken into account.

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70. Learning Innate Face Preferences

James A. Bednar, Risto Miikkulainen

https://doi.org/10.1162/089976603321891792 | pp. 1525-1557

Newborn humans preferentially orient to facelike patterns at birth, but months of experience with faces are required for full face processing abilities to develop. Several models have been proposed for how the interaction of genetic and environmental influences can explain these data. These models generally assume that the brain areas responsible for newborn orienting responses are not capable of learning and are physically separate from those that later learn from real faces. However, it has been difficult to reconcile these models with recent discoveries of face learning in newborns and young infants. We propose a general mechanism by which genetically specified and environment-driven preferences can coexist in the same visual areas. In particular, newborn face orienting may be the result of prenatal exposure of a learning system to internally generated input patterns, such as those found in PGO waves during REM sleep. Simulating this process with the HLISSOM biological model of the visual system, we demonstrate that the combination of learning and internal patterns is an efficient way to specify and develop circuitry for face perception. This prenatal learning can account for the newborn preferences for schematic and photographic images of faces, providing a computational explanation for how genetic influences interact with experience to construct a complex adaptive system.

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71. Learning Optimized Features for Hierarchical Models of Invariant Object Recognition

Heiko Wersing, Edgar Körner

https://doi.org/10.1162/089976603321891800 | pp. 1559-1588

There is an ongoing debate over the capabilities of hierarchical neural feedforward architectures for performing real-world invariant object recognition. Although a variety of hierarchical models exists, appropriate supervised and unsupervised learning methods are still an issue of intense research. We propose a feedforward model for recognition that shares components like weight sharing, pooling stages, and competitive nonlinearities with earlier approaches but focuses on new methods for learning optimal feature-detecting cells in intermediate stages of the hierarchical network. We show that principles of sparse coding, which were previously mostly applied to the initial feature detection stages, can also be employed to obtain optimized intermediate complex features. We suggest a new approach to optimize the learning of sparse features under the constraints of a weight-sharing or convolutional architecture that uses pooling operations to achieve gradual invariance in the feature hierarchy. The approach explicitly enforces symmetry constraints like translation invariance on the feature set. This leads to a dimension reduction in the search space of optimal features and allows determining more efficiently the basis representatives, which achieve a sparse decomposition of the input. We analyze the quality of the learned feature

representation by investigating the recognition performance of the resulting hierarchical network on object and face databases. We show that a hierarchy with features learned on a single object data set can also be applied to face recognition without parameter changes and is competitive with other recent machine learning recognition approaches. To investigate the effect of the interplay between sparse coding and processing nonlinearities, we also consider alternative feedforward pooling nonlinearities such as presynaptic maximum selection and sum-of-squares integration. The comparison shows that a combination of strong competitive nonlinearities with sparse coding offers the best recognition performance in the difficult scenario of segmentation-free recognition in cluttered surround. We demonstrate that for both learning and recognition, a precise segmentation of the objects is not necessary.

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72. Soft Learning Vector Quantization

Sambu Seo, Klaus Obermayer

https://doi.org/10.1162/089976603321891819 | pp. 1589-1604

Learning vector quantization (LVQ) is a popular class of adaptive nearest prototype classifiers for multiclass classification, but learning algorithms from this family have so far been proposed on heuristic grounds. Here, we take a more principled approach and derive two variants of LVQ using a gaussian mixture ansatz. We propose an objective function based on a likelihood ratio and derive a learning rule using gradient descent. The new approach provides a way to extend the algorithms of the LVQ family to different distance measure and allows for the design of "soft" LVQ algorithms. Benchmark results show that the new methods lead to better classification performance than LVQ 2.1. An additional benefit of the new method is that model assumptions are made explicit, so that the method can be adapted more easily to different kinds of problems.

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73. An Efficient Approximation Algorithm for Finding a Maximum Clique Using Hopfield Network Learning

Rong■Long Wang, Zheng Tang, Qi Cao

https://doi.org/10.1162/089976603321891828 | pp. 1605-1619

In this article, we present a solution to the maximum clique problem using a gradient-ascent learning algorithm of the Hopfield neural network. This method provides a near-optimum parallel algorithm for finding a maximum clique. To do this, we use the Hopfield neural network to generate a near-maximum clique and then modify weights in a gradient-ascent direction to allow the network to escape from the state of near-maximum clique to maximum clique or better. The proposed parallel algorithm is tested on two types of random graphs and some benchmark graphs from the Center for Discrete Mathematics and Theoretical Computer Science (DIMACS). The simulation results show that the proposed learning algorithm can find good solutions in reasonable computation time.

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74. The Effect of Noise on a Class of Energy-Based Learning Rules

Armando Bazzani, Daniel Remondini, Nathan Intrator, Gastone Castellani

https://doi.org/10.1162/089976603321891837 | pp. 1621-1640

Westudy the selectivity properties of neurons based on BCM and kurtosis energy functions in a general case of noisy high-dimensional input space. The proposed approach, which is used for characterization of the stable states, can be generalized to a whole class of energy functions. We characterize the critical noise levels beyond which the selectivity is destroyed. We also perform a quantitative analysis of such transitions, which shows interesting dependency on data set size. We observe that the robustness to noise of the BCM neuron (Bienenstock, Cooper, & Munro, 1982; Intrator & Cooper, 1992) increases as a function of dimensionality. We explicitly compute the separability limit of BCM and kurtosis learning rules in the case of a bimodal input distribution. Numerical simulations show a stronger robustness of the BCM rule for practical data set size when compared with kurtosis.

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75. Approximation by Fully Complex Multilayer Perceptrons

Taehwan Kim, Tülay Adal■

https://doi.org/10.1162/089976603321891846 | pp. 1641-1666

We investigate the approximation ability of a multilayer perceptron (MLP) network when it is extended to the complex domain. The main challenge for processing complex data with neural networks has been the lack of bounded and analytic complex nonlinear activation functions in the complex domain, as stated by Liouville's theorem. To avoid the conflict between the boundedness and the analyticity of a nonlinear complex function in the complex domain, a number of ad hoc MLPs that include using two real-valued MLPs, one processing the real part and the other processing the imaginary part, have been traditionally employed. However, since nonanalytic functions do not meet the Cauchy-Riemann conditions, they render themselves into degenerative backpropagation algorithms that compromise the efficiency of nonlinear approximation and learning in the complex vector field. A number of elementary transcendental functions (ETFs) derivable from the entire exponential function e(z) that are analytic are defined as fully complex activation functions and are shown to provide a parsimonious structure for processing data in the complex domain and address most of the shortcomings of the traditional approach. The introduction of ETFs, however, raises a new question in the approximation capability of this fully complex MLP. In this letter, three proofs of the approximation capability of the fully complex MLP are provided based on the characteristics of singularity among ETFs. First, the fully complex MLPs with continuous ETFs over a compact set in the complex vector field are shown to be the universal approximator of any continuous complex mappings. The complex universal approximation theorem extends to bounded measurable ETFs possessing a removable singularity. Finally, it is shown that the output of complex MLPs using ETFs with isolated and essential singularities uniformly converges to any nonlinear mapping in the deleted annulus of singularity nearest to the origin.

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76. Asymptotic Behaviors of Support Vector Machines with Gaussian Kernel

S. Sathiya Keerthi, Chih■Jen Lin

https://doi.org/10.1162/089976603321891855 | pp. 1667-1689

Support vector machines (SVMs) with the gaussian (RBF) kernel have been popular for practical use. Model selection in this class of SVMs involves two hyperparameters: the penalty parameter C and the kernel width sigma. This letter analyzes the behavior of the SVM classifier when these hyperparameters take very small or very large values. Our results help in understanding the hyperparameter space that leads to an efficient heuristic method of searching for hyperparameter values with small generalization errors. The analysis also

indicates that if complete model selection using the gaussian kernel has been conducted, there is no need to consider linear SVM.

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77. Comparison of Model Selection for Regression

Vladimir Cherkassky, Yunqian Ma

https://doi.org/10.1162/089976603321891864 | pp. 1691-1714

We discuss empirical comparison of analytical methods for model selection. Currently, there is no consensus on the best method for finite-sample estimation problems, even for the simple case of linear estimators. This article presents empirical comparisons between classical statistical methods - Akaike information criterion (AIC) and Bayesian information criterion (BIC) - and the structural risk minimization (SRM) method, based on Vapnik-Chervonenkis (VC) theory, for regression problems. Our study is motivated by empirical comparisons in Hastie, Tibshirani, and Friedman (2001), which claims that the SRM method performs poorly for model selection and suggests that AIC yields superior predictive performance. Hence, we present empirical comparisons for various data sets and different types of estimators (linear, subset selection, and k-nearest neighbor regression). Our results demonstrate the practical advantages of VC-based model selection; it consistently outperforms AIC for all data sets. In our study, SRM and BIC methods show similar predictive performance. This discrepancy (between empirical results obtained using the same data) is caused by methodological drawbacks in Hastie et al. (2001), especially in their loose interpretation and application of SRM method. Hence, we discuss methodological issues important for meaningful comparisons and practical application of SRM method. We also point out the importance of accurate estimation of model complexity (VC-dimension) for empirical comparisons and propose a new practical estimate of model complexity for k-nearest neighbors regression.

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78. Computation in a Single Neuron: Hodgkin and Huxley Revisited

Blaise Agüera y Arcas, Adrienne L. Fairhall, William Bialek

https://doi.org/10.1162/08997660360675017 | pp. 1715-1749

A spiking neuron "computes" by transforming a complex dynamical input into a train of action potentials, or spikes. The computation performed by the neuron can be formulated as dimensional reduction, or feature detection, followed by a nonlinear decision function over the low-dimensional space. Generalizations of the reverse correlation technique with white noise input provide a numerical strategy for extracting the relevant low-dimensional features from experimental data, and information theory can be used to evaluate the quality of the low-dimensional approximation. We apply these methods to analyze the simplest biophysically realistic model neuron, the Hodgkin-Huxley (HH) model, using this system to illustrate the general methodological issues. We focus on the features in the stimulus that trigger a spike, explicitly eliminating the effects of interactions between spikes. One can approximate this triggering "feature space" as a two-dimensional linear subspace in the high-dimensional space of input histories, capturing in this way a substantial fraction of the mutual information between inputs and spike time. We find that an even better approximation, however, is to describe the relevant subspace as two dimensional but curved; in this way, we can capture 90% of the mutual information even at high time resolution. Our analysis provides a new understanding of the computational properties of the HH model. While it is common to approximate neural behavior as "integrate and fire," the HH model is not an integrator nor is it well described by a single

threshold.

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79. Learning the Nonlinearity of Neurons from Natural Visual Stimuli

Christoph Kayser, Konrad P. Körding, Peter König

https://doi.org/10.1162/08997660360675026 | pp. 1751-1759

Learning in neural networks is usually applied to parameters related to linear kernels and keeps the nonlinearity of the model fixed. Thus, for successful models, properties and parameters of the nonlinearity have to be specified using a priori knowledge, which often is missing. Here, we investigate adapting the nonlinearity simultaneously with the linear kernel. We use natural visual stimuli for training a simple model of the visual system. Many of the neurons converge to an energy detector matching existing models of complex cells. The overall distribution of the parameter describing the nonlinearity well matches recent physiological results. Controls with randomly shuffled natural stimuli and pink noise demonstrate that the match of simulation and experimental results depends on the higher-order statistical properties of natural stimuli.

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80. Analytic Expressions for Rate and CV of a Type I Neuron Driven by White Gaussian Noise

Benjamin Lindner, André Longtin, Adi R. Bulsara

https://doi.org/10.1162/08997660360675035 | pp. 1761-1788

We study the one-dimensional normal form of a saddle-node system under the influence of additive gaussian white noise and a static "bias current" input parameter, a model that can be looked upon as the simplest version of a type I neuron with stochastic input. This is in contrast with the numerous studies devoted to the noise-driven leaky integrate-and-fire neuron. We focus on the firing rate and coefficient of variation (CV) of the interspike interval density, for which scaling relations with respect to the input parameter and noise intensity are derived. Quadrature formulas for rate and CV are numerically evaluated and compared to numerical simulations of the system and to various approximation formulas obtained in different limiting cases of the model. We also show that caution must be used to extend these results to the Theta neuron model with multiplicative gaussian white noise. The correspondence between the first passage time statistics for the saddle-node model and the Theta neuron model is obtained only in the Stratonovich interpretation of the stochastic Theta neuron model, while previous results have focused only on the Ito interpretation. The correct Stratonovich interpretation yields CVs that are still relatively high, although smaller than in the Ito interpretation; it also produces certain qualitative differences, especially at larger noise intensities. Our analysis provides useful relations for assessing the distance to threshold and the level of synaptic noise in real type I neurons from their firing statistics. We also briefly discuss the effect of finite boundaries (finite values of threshold and reset) on the firing statistics.

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81. What Causes a Neuron to Spike?

Blaise Agüera y Arcas, Adrienne L. Fairhall

https://doi.org/10.1162/08997660360675044 | pp. 1789-1807

The computation performed by a neuron can be formulated as a combination of dimensional reduction in stimulus space and the nonlinearity inherent in a spiking output. White noise stimulus and reverse correlation (the spike-triggered average and spike-triggered covariance) are often used in experimental neuroscience to "ask" neurons which dimensions in stimulus space they are sensitive to and to characterize the nonlinearity of the response. In this article, we apply reverse correlation to the simplest model neuron with temporal dynamics-the leaky integrate-and-fire model-and find that for even this simple case, standard techniques do not recover the known neural computation. To overcome this, we develop novel reverse-correlation techniques by selectively analyzing only "isolated" spikes and taking explicit account of the extended silences that precede these isolated spikes. We discuss the implications of our methods to the characterization of neural adaptation. Although these methods are developed in the context of the leaky integrate-and-fire model, our findings are relevant for the analysis of spike trains from real neurons.

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82. Rate Models for Conductance-Based Cortical Neuronal Networks

Oren Shriki, David Hansel, Haim Sompolinsky

https://doi.org/10.1162/08997660360675053 | pp. 1809-1841

Population rate models provide powerful tools for investigating the principles that underlie the cooperative function of large neuronal systems. However, biophysical interpretations of these models have been ambiguous. Hence, their applicability to real neuronal systems and their experimental validation have been severely limited. In this work, we show that conductance-based models of large cortical neuronal networks can be described by simplified rate models, provided that the network state does not possess a high degree of synchrony. We first derive a precise mapping between the parameters of the rate equations and those of the conductance-based network models for time-independent inputs. This mapping is based on the assumption that the effect of increasing the cell's input conductance on its f-I curve is mainly subtractive. This assumption is confirmed by a single compartment Hodgkin-Huxley type model with a transient potassium A-current. This approach is applied to the study of a network model of a hypercolumn in primary visual cortex. We also explore extensions of the rate model to the dynamic domain by studying the firing-rate response of our conductance-based neuron to time-dependent noisy inputs. We show that the dynamics of this response can be approximated by a time-dependent second-order differential equation. This phenomenological single-cell rate model is used to calculate the response of a conductance-based network to time-dependent inputs.

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83. Neural Representation of Probabilistic Information

Michael Barber, J. W. Clark, Charles H. Anderson

https://doi.org/10.1162/08997660360675062 | pp. 1843-1864

It has been proposed that populations of neurons process information in terms of probability density functions (PDFs) of analog variables. Such analog variables range, for example, from target luminance and depth on the sensory interface to eye position and joint angles on the motor output side. The requirement that analog variables must be processed leads inevitably to a probabilistic description, while the limited precision and lifetime of the neuronal processing units lead naturally to a population representation of information. We show how a time-dependent probability density rho(x; t) over variable x, residing in a specified function space of dimension D, may be decoded from the neuronal activities in a population as a

linear combination of certain decoding functions phi(i)(x), with coefficients given by the N firing rates a(i)(t) (generally with D << N). We show how the neuronal encoding process may be described by projecting a set of complementary encoding functions phi;(i)(x) on the probability density rho(x; t), and passing the result through a rectifying nonlinear activation function. We show how both encoders phi;(i)(x) and decoders phi(i)(x) may be determined by minimizing cost functions that quantify the inaccuracy of the representation. Expressing a given computation in terms of manipulation and transformation of probabilities, we show how this representation leads to a neural circuit that can carry out the required computation within a consistent Bayesian framework, with the synaptic weights being explicitly generated in terms of encoders, decoders, conditional probabilities, and priors.

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84. Learning the Gestalt Rule of Collinearity from Object Motion

Carsten Prodöhl, Rolf P. Würtz, Christoph von der Malsburg

https://doi.org/10.1162/08997660360675071 | pp. 1865-1896

The Gestalt principle of collinearity (and curvilinearity) is widely regarded as being mediated by the long-range connection structure in primary visual cortex. We review the neurophysiological and psychophysical literature to argue that these connections are developed from visual experience after birth, relying on coherent object motion. We then present a neural network model that learns these connections in an unsupervised Hebbian fashion with input from real camera sequences. The model uses spatiotemporal retinal filtering, which is very sensitive to changes in the visual input. We show that it is crucial for successful learning to use the correlation of the transient responses instead of the sustained ones. As a consequence, learning works best with video sequences of moving objects. The model addresses a special case of the fundamental question of what represents the necessary a priori knowledge the brain is equipped with at birth so that the self-organized process of structuring by experience can be successful.

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85. Recurrent Neural Networks with Small Weights Implement Definite Memory Machines

Barbara Hammer, Peter Ti■o

https://doi.org/10.1162/08997660360675080 | pp. 1897-1929

Recent experimental studies indicate that recurrent neural networks initialized with “small” weights are inherently biased toward definite memory machines (Ti■no, ■er■anský, & Be■ušková, 2002a, 2002b). This article establishes a theoretical counterpart: transition function of recurrent network with small weights and squashing activation function is a contraction. We prove that recurrent networks with contractive transition function can be approximated arbitrarily well on input sequences of unbounded length by a definite memory machine. Conversely, every definite memory machine can be simulated by a recurrent network with contractive transition function. Hence, initialization with small weights induces an architectural bias into learning with recurrent neural networks. This bias might have benefits from the point of view of statistical learning theory: it emphasizes one possible region of the weight space where generalization ability can be formally proved. It is well known that standard recurrent neural networks are not distribution independent learnable in the probably approximately correct (PAC) sense if arbitrary precision and inputs are considered. We prove that recurrent networks with contractive transition function with a fixed contraction parameter fulfill the so-called distribution independent uniform convergence of empirical distances property and hence,

unlike general recurrent networks, are distribution independent PAC learnable.

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86. Architectural Bias in Recurrent Neural Networks: Fractal Analysis

Peter Ti■o, Barbara Hammer

https://doi.org/10.1162/08997660360675099 | pp. 1931-1957

We have recently shown that when initialized with “small” weights, recurrent neural networks (RNNs) with standard sigmoid-type activation functions are inherently biased toward Markov models; even prior to any training, RNN dynamics can be readily used to extract finite memory machines (Hammer & Ti■o, 2002; Ti■o, ■er■anský, &Be■ušková, 2002a, 2002b). Following Christiansen and Chater (1999), we refer to this phenomenon as the architectural bias of RNNs. In this article, we extend our work on the architectural bias in RNNs by performing a rigorous fractal analysis of recurrent activation patterns. We assume the network is driven by sequences obtained by traversing an underlying finite-state transition diagram&a scenario that has been frequently considered in the past, for example, when studying RNN-based learning and implementation of regular grammars and finite-state transducers. We obtain lower and upper bounds on various types of fractal dimensions, such as box counting and Hausdorff dimensions. It turns out that not only can the recurrent activations inside RNNs with small initial weights be explored to build Markovian predictive models, but also the activations form fractal clusters, the dimension of which can be bounded by the scaled entropy of the underlying driving source. The scaling factors are fixed and are given by the RNN parameters.

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87. An Effective Bayesian Neural Network Classifier with a Comparison Study to Support Vector Machine

Faming Liang

https://doi.org/10.1162/08997660360675107 | pp. 1959-1989

We propose a new Bayesian neural network classifier, different from that commonly used in several respects, including the likelihood function, prior specification, and network structure. Under regularity conditions, we show that the decision boundary determined by the new classifier will converge to the true one. We also propose a systematic implementation for the new classifier. In our implementation, the tune of connection weights, the selection of hidden units, and the selection of input variables are unified by sampling from the joint posterior distribution of the network structure and connection weights. The numerical results show that the new classifier consistently outperforms the commonly used Bayesian neural network classifier and the support vector machine in terms of generalization performance. The reason for the inferiority of the commonly used Bayesian neural network classifier and the support vector machine is discussed at length.

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88. Variational Bayesian Learning of ICA with Missing Data

Kwokleung Chan, Te-Won Lee, Terrence J. Sejnowski

https://doi.org/10.1162/08997660360675116 | pp. 1991-2011

Missing data are common in real-world data sets and are a problem for many estimation techniques. We have developed a variational Bayesian method to perform independent component analysis (ICA) on

high-dimensional data containing missing entries. Missing data are handled naturally in the Bayesian framework by integrating the generative density model. Modeling the distributions of the independent sources with mixture of gaussians allows sources to be estimated with different kurtosis and skewness. Unlike the maximum likelihood approach, the variational Bayesian method automatically determines the dimensionality of the data and yields an accurate density model for the observed data without overfitting problems. The technique is also extended to the clusters of ICA and supervised classification framework.

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89. Have Brain Dynamics Evolved? Should We Look for Unique Dynamics in the Sapient Species?

Theodore H. Bullock

https://doi.org/10.1162/089976603322297269 | pp. 2013-2027

Ongoing "spontaneous" electrical field potentials of assemblies of neurons in the brains of diverse animal groups differ widely in character and amplitude without obvious explanation. There may be correlates with other measures of brain complexity, such as histological differentiation, but so far there are no known differences between the EEGs of humans and other mammals or between mammals and reptiles, amphibians or fish, apart from amplitude. The proposition is defended that further search for descriptors or statistical, probably nonlinear features of the time series will reveal consistent differences, meaning that we have so far missed major features of the natural history of EEGs, just as we have thus far relatively neglected the identification of features of the physiology of the brain relevant to its evolution of complexity through major grades of phyla, classes, and orders.

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90. Is There Something Out There? Inferring Space from Sensorimotor Dependencies

David Philipona, J. Kevin O'Regan, Jean■Pierre Nadal

https://doi.org/10.1162/089976603322297278 | pp. 2029-2049

This letter suggests that in biological organisms, the perceived structure of reality, in particular the notions of body, environment, space, object, and attribute, could be a consequence of an effort on the part of brains to account for the dependency between their inputs and their outputs in terms of a small number of parameters. To validate this idea, a procedure is demonstrated whereby the brain of a (simulated) organism with arbitrary input and output connectivity can deduce the dimensionality of the rigid group of the space underlying its input-output relationship, that is, the dimension of what the organism will call physical space.

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91. A Developmental Approach Aids Motor Learning

V. Ivanchenko, Robert A. Jacobs

https://doi.org/10.1162/089976603322297287 | pp. 2051-2065

Bernstein (1967) suggested that people attempting to learn to perform a difficult motor task try to ameliorate the degrees-of-freedom problem through the use of a developmental progression. Early in training, people maintain a subset of their control parameters (e.g., joint positions) at constant settings and attempt to learn to perform the task by varying the values of the remaining parameters. With practice, people refine and

improve this early-learned control strategy by also varying those parameters that were initially held constant. We evaluated Bernstein's proposed developmental progression using six neural network systems and found that a network whose training included developmental progressions of both its trajectory and its feedback gains outperformed all other systems. These progressions, however, yielded performance benefits only on motor tasks that were relatively difficult to learn. We conclude that development can indeed aid motor learning.

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92. Cell Responsiveness in Macaque Superior Temporal Polysensory Area Measured by Temporal Discriminants

Jessica A. Turner, Kathleen Anderson, Ralph M. Siegel

https://doi.org/10.1162/089976603322297296 | pp. 2067-2090

The firing-rate data from 341 cells from two macaques' superion temporal polysensory area (STPa) were subjected to three different analyses to determine the temporal firing-rate patterns in response to visual optic flow patterns. The data were collected while the monkey viewed four types of optic flow and responded to the change in the display. The mean firing rate (MFR) analysis considered the mean change in firing rate for 500 ms after stimulus onset; the discriminant (DIS) analysis and the principal components (PCA+DIS) analysis considered the change in time-binned firing rate over 1000 ms after stimulus presentation, using bin sizes of 30 to 500 ms. The DIS analysis used a step-down discriminant analysis to find temporal windows in which the cell's firing rate could discriminate among the stimuli; the PCA+DIS analysis extracted the principal components of the cell's firing rates without regard for the stimulus type and then applied a step-down discriminant analysis to the PCA scores to determine whether any of the principal components could discriminate among the stimuli. The two temporal analyses found cells sensitive to the optic flows that the MFR analysis missed. A small proportion of cells showed multiple selectivities under the temporal analyses. Thus, the temporal analyses give a more complete representation of the information encoded by the firing properties of STPa neurons. Finally, this approach incorporates temporal approaches with classical statistical techniques in order to select tuned neurons from a population in an unbiased manner.

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93. Local Interactions in Neural Networks Explain Global Effects in Gestalt Processing and Masking

Michael H. Herzog, Udo Ernst, Axel Etzold, Christian W. Eurich

https://doi.org/10.1162/089976603322297304 | pp. 2091-2113

One of the fundamental and puzzling questions in vision research is how objects are segmented from their backgrounds and how object formation evolves in time. The recently discovered shine-through effect allows one to study object segmentation and object formation of a masked target depending on the spatiotemporal Gestalt of the masking stimulus (Herzog & Koch, 2001). In the shine-through effect, a vernier (two abutting lines) precedes a grating for a very short time. For small gratings, the vernier remains invisible while it regains visibility as a shine-through element for extended and homogeneous gratings. However, even subtle deviations from the homogeneity of the grating diminish or even abolish shine-through. At first glance, these results suggest that explanations of these effects have to rely on high-level Gestalt terminology such as homogeneity rather than on low-level properties such as luminance (Herzog, Fahle, & Koch, 2001). Here, we show that a simple neural network model of the Wilson-Cowan type qualitatively and quantitatively

explains the basic effects in the shine-through paradigm, although the model does not contain any explicit, global Gestalt processing. Visibility of the target vernier corresponds to transient activation of neural populations resulting from the dynamics of local lateral interactions of excitatory and inhibitory layers of neural populations.

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94. Computing with Populations of Monotonically Tuned Neurons

Emmanuel Guigon

https://doi.org/10.1162/089976603322297313 | pp. 2115-2127

The parametric variation in neuronal discharge according to the values of sensory or motor variables strongly influences the collective behavior of neuronal populations. A multitude of studies on the populations of broadly tuned neurons (e.g., cosine tuning) have led to such well-known computational principles as population coding, noise suppression, and line attractors. Much less is known about the properties of populations of monotonically tuned neurons. In this letter, we show that there exists an efficient weakly biased linear estimator for monotonic populations and that neural processing based on linear collective computation and least-square error learning in populations of intensity-coded neurons has specific generalization capacities.

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95. A Simple and Stable Numerical Solution for the Population Density Equation

Marc de Kamps

https://doi.org/10.1162/089976603322297322 | pp. 2129-2146

A population density description of large populations of neurons has generated considerable interest recently. The evolution in time of the population density is determined by a partial differential equation (PDE). Most of the algorithms proposed to solve this PDE have used finite difference schemes. Here, I use the method of characteristics to reduce the PDE to a set of ordinary differential equations, which are easy to solve. The method is applied to leaky-integrate-and-fire neurons and produces an algorithm that is efficient and yields a stable and manifestly nonnegative density. Contrary to algorithms based directly on finite difference schemes, this algorithm is insensitive to large density gradients, which may occur during evolution of the density.

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96. Slow Feature Analysis: A Theoretical Analysis of Optimal Free Responses

Laurenz Wiskott

https://doi.org/10.1162/089976603322297331 | pp. 2147-2177

Temporal slowness is a learning principle that allows learning of invariant representations by extracting slowly varying features from quickly varying input signals. Slow feature analysis (SFA) is an efficient algorithm based on this principle and has been applied to the learning of translation, scale, and other invariances in a simple model of the visual system. Here, a theoretical analysis of the optimization problem solved by SFA is presented, which provides a deeper understanding of the simulation results obtained in previous studies.

97. Synchrony of Fast-Spiking Interneurons Interconnected by GABAergic and Electrical Synapses

Masaki Nomura, Tomoki Fukai, Toshio Aoyagi

https://doi.org/10.1162/089976603322297340 | pp. 2179-2198

Fast-spiking (FS) interneurons have specific types (Kv3.1/3.2 type) of the delayed potassium channel, which differ from the conventional Hodgkin-Huxley (HH) type potassium channel (Kv1.3 type) in several aspects. In this study, we show dramatic effects of the Kv3.1/3.2 potassium channel on the synchronization of the FS interneurons. We show analytically that two identical electrically coupled FS interneurons modeled with Kv3.1/3.2 channel fire synchronously at arbitrary firing frequencies, unlike similarly coupled FS neurons modeled with Kv1.3 channel that show frequency-dependent synchronous and antisynchronous firing states. Introducing GABA(A) receptor-mediated synaptic connections into an FS neuron pair tends to induce an antisynchronous firing state, even if the chemical synapses are bidirectional. Accordingly, an FS neuron pair connected simultaneously by electrical and chemical synapses achieves both synchronous firing state and antisynchronous firing state in a physiologically plausible range of the conductance ratio between electrical and chemical synapses. Moreover, we find that a large-scale network of FS interneurons connected by gap junctions and bidirectional GABAergic synapses shows similar bistability in the range of gamma frequencies (30-70 Hz).

98. Activation Functions Defined on Higher-Dimensional Spaces for Approximation on Compact Sets with and without Scaling

Yoshifusa Ito

https://doi.org/10.1162/089976603322297359 | pp. 2199-2226

Let g be a slowly increasing function of locally bounded variation defined on R c , 1 ≤c≤d. We investigate when g can be an activation function of the hidden-layer units of three-layer neural networks that approximate continuous functions on compact sets. If the support of the Fourier transform of g includes a converging sequence of points with distinct distances from the origin, it can be an activation function without scaling. If and only if the support of its Fourier transform includes a point other than the origin, it can be an activation function with scaling. We also look for a condition on which an activation function can be used for approximation without rotation. Any nonpolynomial functions can be activation functions with scaling, and many familiar functions, such as sigmoid functions and radial basis functions, can be activation functions without scaling. With or without scaling, some of them defined on R d can be used without rotation even if they are not spherically symmetric.

99. Bayesian Trigonometric Support Vector Classifier

Wei Chu, S. Sathiya Keerthi, Chong Jin Ong

https://doi.org/10.1162/089976603322297368 | pp. 2227-2254

This letter describes Bayesian techniques for support vector classification. In particular, we propose a novel differentiable loss function, called the trigonometric loss function, which has the desirable characteristic of natural normalization in the likelihood function, and then follow standard gaussian processes techniques to

set up a Bayesian framework. In this framework, Bayesian inference is used to implement model adaptation, while keeping the merits of support vector classifier, such as sparseness and convex programming. This differs from standard gaussian processes for classification. Moreover, we put forward class probability in making predictions. Experimental results on benchmark data sets indicate the usefulness of this approach.

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100. Doubly Distributional Population Codes: Simultaneous Representation of Uncertainty and Multiplicity

Maneesh Sahani, Peter Dayan

https://doi.org/10.1162/089976603322362356 | pp. 2255-2279

Perceptual inference fundamentally involves uncertainty, arising from noise in sensation and the ill-posed nature of many perceptual problems. Accurate perception requires that this uncertainty be correctly represented, manipulated, and learned about. The choices subjects make in various psychophysical experiments suggest that they do indeed take such uncertainty into account when making perceptual inferences, posing the question as to how uncertainty is represented in the activities of neuronal populations. Most theoretical investigations of population coding have ignored this issue altogether; the few existing proposals that address it do so in such a way that it is fatally conflated with another facet of perceptual problems that also needs correct handling: multiplicity (that is, the simultaneous presence of multiple distinct stimuli). We present and validate a more powerful proposal for the way that population activity may encode uncertainty, both distinctly from and simultaneously with multiplicity.

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101. Firing Rate of the Noisy Quadratic Integrate-and-Fire Neuron

Nicolas Brunel, Peter E. Latham

https://doi.org/10.1162/089976603322362365 | pp. 2281-2306

We calculate the firing rate of the quadratic integrate-and-fire neuron in response to a colored noise input current. Such an input current is a good approximation to the noise due to the random bombardment of spikes, with the correlation time of the noise corresponding to the decay time of the synapses. The key parameter that determines the firing rate is the ratio of the correlation time of the colored noise, tau(s), to the neuronal time constant, tau(m). We calculate the firing rate exactly in two limits: when the ratio, tau(s)/tau(m), goes to zero (white noise) and when it goes to infinity. The correction to the short correlation time limit is O(tau(s)/tau(m)), which is qualita tively different from that of the leaky integrate-and-fire neuron, where the correction is O(radical tau(s)/tau(m)). The difference is due to the different boundary conditions of the probability density function of the membrane potential of the neuron at firing threshold. The correction to the long correlation time limit is O(tau(m)/tau(s)). By combining the short and long correlation time limits, we derive an expression that provides a good approximation to the firing rate over the whole range of tau(s)/tau(m) in the suprathreshold regime-that is, in a regime in which the average current is sufficient to make the cell fire. In the subthreshold regime, the expression breaks down somewhat when tau(s) becomes large compared to tau(m).

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102. Pattern Filtering for Detection of Neural Activity, with Examples from HVc Activity During Sleep in Zebra Finches

Zhiyi Chi, Peter L. Rauske, Daniel Margoliash

https://doi.org/10.1162/089976603322362374 | pp. 2307-2337

The detection of patterned spiking activity is important in the study of neural coding. A pattern filtering approach is developed for pattern detection under the framework of point processes, which offers flexibility in combining temporal details and firing rates. The detection combines multiple steps of filtering in a coarse-to-fine manner. Under some conditional Poisson assumptions on the spiking activity, each filtering step is equivalent to classifying by likelihood ratios all the data segments as targets or as background sequences. Unlike previous studies, where global surrogate data were used to evaluate the statistical significance of the detected patterns, a localized p-test procedure is developed, which better accounts for firing modulation and nonstationarity in spiking activity. Common temporal structures of patterned activity are learned using an entropy-based alignment procedure, without relying on metrics or pair-wise alignment. Applications of pattern filtering to single, presumptive interneurons recorded in the nucleus HVc of zebra finch are illustrated. These demonstrate a match between the auditory-evoked response to playback of the individual bird's own song and spontaneous activity during sleep. Small temporal compression or expansion, or both, is required for optimal matching of spontaneous patterns to stimulus-evoked activity.

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103. Synaptic Depression Leads to Nonmonotonic Frequency Dependence in the Coincidence Detector

Shawn Mikula, Ernst Niebur

https://doi.org/10.1162/089976603322362383 | pp. 2339-2358

In this letter, we extend our previous analytical results (Mikula & Niebur, 2003) for the coincidence detector by taking into account probabilistic frequency-dependent synaptic depression. We present a solution for the steady-state output rate of an ideal coincidence detector receiving an arbitrary number of input spike trains with identical binomial count distributions (which includes Poisson statistics as a special case) and identical arbitrary pairwise cross-correlations, from zero correlation (independent processes) to perfect correlation (identical processes). Synapses vary their efficacy probabilistically according to the observed depression mechanisms. Our results show that synaptic depression, if made sufficiently strong, will result in an inverted U-shaped curve for the output rate of a coincidence detector as a function of input rate. This leads to the counterintuitive prediction that higher presynaptic (input) rates may lead to lower postsynaptic (output) rates where the output rate may fall faster than the inverse of the input rate.

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104. Probing Changes in Neural Interaction During Adaptation

Li Qiang Zhu, Ying■Cheng Lai, Frank C. Hoppensteadt, Jiping He

https://doi.org/10.1162/089976603322362392 | pp. 2359-2377

A procedure is developed to probe the changes in the functional interactions among neurons in primary motor cortex of the monkey brain during adaptation. A monkey is trained to learn a new skill, moving its arm to reach a target under the influence of external perturbations. The spike trains of multiple neurons in the primary motor cortex are recorded simultaneously. We utilize the methodology of directed transfer function,

derived from a class of linear stochastic models, to quantify the causal interactions between the neurons. We find that the coupling between the motor neurons tends to increase during the adaptation but return to the original level after the adaptation. Furthermore, there is evidence that adaptation tends to affect the topology of the neural network, despite the approximate conservation of the average coupling strength in the network before and after the adaptation.

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105. Memory Encoding by Theta Phase Precession in the Hippocampal Network

Naoyuki Sato, Yoko Yamaguchi

https://doi.org/10.1162/089976603322362400 | pp. 2379-2397

Recent experimental evidence on spike-timing-dependent plasticity and on phase precession (i.e., the theta rhythm dependent firing of rat hippocampal cells) associates the contribution of phase precession to episodic memory. This article aims at clarifying the role of phase precession in memory storage. Computer simulations show that the memory storage in the behavioral timescale varies in timescale of the temporal sequence from half a second to several seconds. In contrast, the memory storage caused by traditional rate coding is restricted to the temporal sequence within 40 ms. During phase precession, memory storage of a single trial experience is possible, even in the presence of noise. It is therefore concluded that encoding by phase precession is appropriate for memory storage of the temporal sequence in the behavioral timescale.

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106. A Computational Model as Neurodecoder Based on Synchronous Oscillation in the Visual Cortex

Zhao Songnian, Xiaoyun Xiong, Yao Guozheng, Fu Zhi

https://doi.org/10.1162/089976603322362419 | pp. 2399-2418

Based on synchronized responses of neuronal populations in the visual cortex to external stimuli, we proposed a computational model consisting primarily of a neuronal phase-locked loop (NPLL) and multiscaled operator. The former reveals the function of synchronous oscillations in the visual cortex. Regardless of which of these patterns of the spike trains may be an average firing-rate code, a spike-timing code, or a rate-time code, the NPLL can decode original visual information from neuronal spike trains modulated with patterns of external stimuli, because a voltage-controlled oscillator (VCO), which is included in the NPLL, can precisely track neuronal spike trains and instantaneous variations, that is, VCO can make a copy of an external stimulus pattern. The latter, however, describes multi-scaled properties of visual information processing, but not merely edge and contour detection. In this study, in which we combined NPLL with a multiscaled operator and maximum likelihood estimation, we proved that the model, as a neurodecoder, implements optimum algorithm decoding visual information from neuronal spike trains at the system level. At the same time, the model also obtains increasingly important supports, which come from a series of experimental results of neurobiology on stimulus-specific neuronal oscillations or synchronized responses of the neuronal population in the visual cortex. In addition, the problem of how to describe visual acuity and multiresolution of vision by wavelet transform is also discussed. The results indicate that the model provides a deeper understanding of the role of synchronized responses in decoding visual information.

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107. Parameter Estimation of Sigmoid Superpositions: Dynamical System Approach

Ivan Tyukin, Cees van Leeuwen, Danil Prokhorov

https://doi.org/10.1162/089976603322362428 | pp. 2419-2455

Superposition of sigmoid function over a finite time interval is shown to be equivalent to the linear combination of the solutions of a linearly parameterized system of logistic differential equations. Due to the linearity with respect to the parameters of the system, it is possible to design an effective procedure for parameter adjustment. Stability properties of this procedure are analyzed.

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108. On the Partitioning Capabilities of Feedforward Neural Networks with Sigmoid Nodes

Konstantinos Koutroumbas

https://doi.org/10.1162/089976603322362437 | pp. 2457-2481

In this letter, the capabilities of feedforward neural networks (FNNs) on the realization and approximation of functions of the form g: R(l) --> A, which partition the R(l) space into polyhedral sets, each one being assigned to one out of the c classes of A, are investigated. More specifically, a constructive proof is given for the fact that FNNs consisting of nodes having sigmoid output functions are capable of approximating any function g with arbitrary accuracy. Also, the capabilities of FNNs consisting of nodes having the hard limiter as output function are reviewed. In both cases, the two-class as well as the multiclass cases are considered.

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109. Dynamical Consequences of Fast-Rising, Slow-Decaying Synapses in Neuronal Networks

Bard Ermentrout

https://doi.org/10.1162/089976603322385054 | pp. 2483-2522

Synapses that rise quickly but have long persistence are shown to have certain computational advantages. They have some unique mathematical properties as well and in some instances can make neurons behave as if they are weakly coupled oscillators. This property allows us to determine their synchronization properties. Furthermore, slowly decaying synapses allow recurrent networks to maintain excitation in the absence of inputs, whereas faster decaying synapses do not. There is an interaction between the synaptic strength and the persistence that allows recurrent networks to fire at low rates if the synapses are sufficiently slow. Waves and localized structures are constructed in spatially extended networks with slowly decaying synapses.

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110. A Universal Model for Spike-Frequency Adaptation

Jan Benda, Andreas V. M. Herz

https://doi.org/10.1162/089976603322385063 | pp. 2523-2564

Spike-frequency adaptation is a prominent feature of neural dynamics. Among other mechanisms, various ionic currents modulating spike generation cause this type of neural adaptation. Prominent examples are

voltage-gated potassium currents (M-type currents), the interplay of calcium currents and intracellular calcium dynamics with calcium-gated potassium channels (AHP-type currents), and the slow recovery from inactivation of the fast sodium current. While recent modeling studies have focused on the effects of specific adaptation currents, we derive a universal model for the firing-frequency dynamics of an adapting neuron that is independent of the specific adaptation process and spike generator. The model is completely defined by the neuron's onset f-I curve, the steady-state f-I curve, and the time constant of adaptation. For a specific neuron, these parameters can be easily determined from electrophysiological measurements without any pharmacological manipulations. At the same time, the simplicity of the model allows one to analyze mathematically how adaptation influences signal processing on the single-neuron level. In particular, we elucidate the specific nature of high-pass filter properties caused by spike-frequency adaptation. The model is limited to firing frequencies higher than the reciprocal adaptation time constant and to moderate fluctuations of the adaptation and the input current. As an extension of the model, we introduce a framework for combining an arbitrary spike generator with a generalized adaptation current.

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111. An Adjustment to the Time-Rescaling Method for Application to Short-Trial Spike Train Data

Matthew C. Wiener

https://doi.org/10.1162/089976603322385072 | pp. 2565-2576

It is important to validate models of neural data using appropriate goodness-of-fit measures. Models summarizing some response features--for example, spike count distributions or peristimulus time histograms--can be assessed using standard statistical tools. Measuring the fit of a full point-process model of spike trains is more difficult. Recently, Barbieri, Quirk, Frank, Wilson, and Brown (2001) and Brown, Barbieri, Ventura, Kass, and Frank (2002) presented a method for rescaling time so that if an underlying description correctly describes the conditional intensity function of a point process, the rescaling will convert the process into a homogeneous Poisson process. The corresponding interevent intervals are exponential with mean 1 and can be transformed to be uniform; tests of the uniformity of the transformed intervals are thus tests of how well the model fits the data. When the lengths of interevent intervals are comparable to the length of the observation window, as can happen in common neurophysiology paradigms using short trials, the fact that long intervals cannot be observed (are censored) can cause the tests based on time rescaling to reject a correct model inappropriately. This article presents a simple adjustment to the time-rescaling method to account for interval censoring, avoiding inappropriate rejection of acceptable models for short-trial data. We illustrate the adjustment's effect using both simulated data and short-trial data from monkey primary visual cortex.

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112. Characterization of Subthreshold Voltage Fluctuations in Neuronal Membranes

Michael Rudolph, Alain Destexhe

https://doi.org/10.1162/089976603322385081 | pp. 2577-2618

Synaptic noise due to intense network activity can have a significant impact on the electrophysiological properties of individual neurons. This is the case for the cerebral cortex, where ongoing activity leads to strong barrages of synaptic inputs, which act as the main source of synaptic noise affecting on neuronal

dynamics. Here, we characterize the subthreshold behavior of neuronal models in which synaptic noise is represented by either additive or multiplicative noise, described by Ornstein-Uhlenbeck processes. We derive and solve the Fokker-Planck equation for this system, which describes the time evolution of the probability density function for the membrane potential. We obtain an analytic expression for the membrane potential distribution at steady state and compare this expression with the subthreshold activity obtained in Hodgkin-Huxley-type models with stochastic synaptic inputs. The differences between multiplicative and additive noise models suggest that multiplicative noise is adequate to describe the high-conductance states similar to in vivo conditions. Because the steady-state membrane potential distribution is easily obtained experimentally, this approach provides a possible method to estimate the mean and variance of synaptic conductances in real neurons.

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113. Modeling Reaching Impairment After Stroke Using a Population Vector Model of Movement Control That Incorporates Neural Firing-Rate Variability

David J. Reinkensmeyer, Mario G. Iobbi, Leonard Kahn, Derek G. Kamper, Craig D. Takahashi
https://doi.org/10.1162/089976603322385090 | pp. 2619-2642

The directional control of reaching after stroke was simulated by including cell death and firing-rate noise in a population vector model of movement control. In this model, cortical activity was assumed to cause the hand to move in the direction of a population vector, defined by a summation of responses from neurons with cosine directional tuning. Two types of directional error were analyzed: the between-target variability, defined as the standard deviation of the directional error across a wide range of target directions, and the within-target variability, defined as the standard deviation of the directional error for many reaches to a single target. Both between- and within-target variability increased with increasing cell death. The increase in between-target variability arose because cell death caused a nonuniform distribution of preferred directions. The increase in within-target variability arose because the magnitude of the population vector decreased more quickly than its standard deviation for increasing cell death, provided appropriate levels of firing-rate noise were present. Comparisons to reaching data from 29 stroke subjects revealed similar increases in between- and within-target variability as clinical impairment severity increased. Relationships between simulated cell death and impairment severity were derived using the between- and within-target variability results. For both relationships, impairment severity increased similarly with decreasing percentage of surviving cells, consistent with results from previous imaging studies. These results demonstrate that a population vector model of movement control that incorporates cosine tuning, linear summation of unitary responses, firing-rate noise, and random cell death can account for some features of impaired arm movement after stroke.

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114. Radius Margin Bounds for Support Vector Machines with the RBF Kernel

Kai-Min Chung, Wei■Chun Kao, Chia■Liang Sun, Li-Lun Wang, Chih■Jen Lin
https://doi.org/10.1162/089976603322385108 | pp. 2643-2681

An important approach for efficient support vector machine (SVM) model selection is to use differentiable bounds of the leave-one-out (loo) error. Past efforts focused on finding tight bounds of loo (e.g., radius margin bounds, span bounds). However, their practical viability is still not very satisfactory. Duan, Keerthi, and Poo (2003) showed that radius margin bound gives good prediction for L2-SVM, one of the cases we

look at. In this letter, through analyses about why this bound performs well for L2-SVM, we show that finding a bound whose minima are in a region with small loo values may be more important than its tightness. Based on this principle, we propose modified radius margin bounds for L1-SVM (the other case) where the original bound is applicable only to the hard-margin case. Our modification for L1-SVM achieves comparable performance to L2-SVM. To study whether L1- or L2-SVM should be used, we analyze other properties, such as their differentiability, number of support vectors, and number of free support vectors. In this aspect, L1-SVM possesses the advantage of having fewer support vectors. Their implementations are also different, so we discuss related issues in detail.

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115. Accurate On-line Support Vector Regression

Junshui Ma, James Theiler, Simon Perkins

https://doi.org/10.1162/089976603322385117 | pp. 2683-2703

Batch implementations of support vector regression (SVR) are inefficient when used in an on-line setting because they must be retrained from scratch every time the training set is modified. Following an incremental support vector classification algorithm introduced by Cauwenberghs and Poggio (2001), we have developed an accurate on-line support vector regression (AOSVR) that efficiently updates a trained SVR function whenever a sample is added to or removed from the training set. The updated SVR function is identical to that produced by a batch algorithm. Applications of AOSVR in both on-line and cross-validation scenarios are presented. In both scenarios, numerical experiments indicate that AOSVR is faster than batch SVR algorithms with both cold and warm start.

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116. Speeding Up Backpropagation Using Multiobjective Evolutionary Algorithms

Hussein A. Abbass

https://doi.org/10.1162/089976603322385126 | pp. 2705-2726

The use of backpropagation for training artificial neural networks (ANNs) is usually associated with a long training process. The user needs to experiment with a number of network architectures; with larger networks, more computational cost in terms of training time is required. The objective of this letter is to present an optimization algorithm, comprising a multiobjective evolutionary algorithm and a gradient-based local search. In the rest of the letter, this is referred to as the memetic Pareto artificial neural network algorithm for training ANNs. The evolutionary approach is used to train the network and simultaneously optimize its architecture. The result is a set of networks, with each network in the set attempting to optimize both the training error and the architecture. We also present a self-adaptive version with lower computational cost. We show empirically that the proposed method is capable of reducing the training time compared to gradient-based techniques.

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117. General-Purpose Computation with Neural Networks: A Survey of Complexity Theoretic Results

Ji■■■ Š■■ma, Pekka Orponen

https://doi.org/10.1162/089976603322518731 | pp. 2727-2778

We survey and summarize the literature on the computational aspects of neural network models by presenting a detailed taxonomy of the various models according to their complexity theoretic characteristics. The criteria of classification include the architecture of the network (feedforward versus recurrent), time model (discrete versus continuous), state type (binary versus analog), weight constraints (symmetric versus asymmetric), network size (finite nets versus infinite families), and computation type (deterministic versus probabilistic), among others. The underlying results concerning the computational power and complexity issues of perceptron, radial basis function, winner-take-all, and spiking neural networks are briefly surveyed, with pointers to the relevant literature. In our survey, we focus mainly on the digital computation whose inputs and outputs are binary in nature, although their values are quite often encoded as analog neuron states. We omit the important learning issues.

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118. Dynamics of Deterministic and Stochastic Paired Excitatory—Inhibitory Delayed Feedback

Carlo R. Laing, André Longtin

https://doi.org/10.1162/089976603322518740 | pp. 2779-2822

We examine the effects of paired delayed excitatory and inhibitory feedback on a single integrate-and-fire neuron with reversal potentials embedded within a feedback network. These effects are studied using bifurcation theory and numerical analysis. The feedback occurs through modulation of the excitatory and inhibitory conductances by the previous firing history of the neuron; as a consequence, the feedback also modifies the membrane time constant. Such paired feedback is ubiquitous in the nervous system. We assume that the feedback dynamics are slower than the membrane time constant, which leads to a rate model formulation. Our article provides an extensive analysis of the possible dynamical behaviors of such simple yet realistic neural loops as a function of the balance between positive and negative feedback, with and without noise, and offers insight into the potential behaviors such loops can exhibit in response to time-varying external inputs. With excitatory feedback, the system can be quiescent, can be periodically firing, or can exhibit bistability between these two states. With inhibitory feedback, quiescence, oscillatory firing rates, and bistability between constant and oscillatory firing-rate solutions are possible. The general case of paired feedback exhibits a blend of the behaviors seen in the extreme cases and can produce chaotic firing. We further derive a condition for a dynamically balanced paired feedback in which there is neither bistability nor oscillations. We also show how a biophysically plausible smoothing of the firing function by noise can modify the existence and stability of fixed points and oscillations of the system. We take advantage in our mathematical analysis of the existence of an invariant manifold, which reduces the dimensionality of the dynamics, and prove the stability of this manifold. The novel computational challenges involved in analyzing such dynamics with and without noise are also described. Our results demonstrate that a paired delayed feedback loop can act as a sophisticated computational unit, capable of switching between a variety of behaviors depending on the input current, the relative strengths and asymmetry of the two parallel feedback pathways, and the delay distributions and noise level.

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119. Differences in Spiking Patterns Among Cortical Neurons

Shigeru Shinomoto, Keisetsu Shima, Jun Tanji

https://doi.org/10.1162/089976603322518759 | pp. 2823-2842

Spike sequences recorded from four cortical areas of an awake behaving monkey were examined to explore characteristics that vary among neurons. We found that a measure of the local variation of interspike intervals, L(V), is nearly the same for every spike sequence for any given neuron, while it varies significantly among neurons. The distributions of L(V) values for neuron ensembles in three of the four areas were found to be distinctly bimodal. Two groups of neurons classified according to the spiking irregularity exhibit different responses to the same stimulus. This suggests that neurons in each area can be classified into different groups possessing unique spiking statistics and corresponding functional properties.

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120. Hybrid Integrate-and-Fire Model of a Bursting Neuron

Barbara J. Breen, William Christopher Gerken, Robert J. Butera

https://doi.org/10.1162/089976603322518768 | pp. 2843-2862

We present a reduction of a Hodgkin-Huxley (HH)--style bursting model to a hybridized integrate-and-fire (IF) formalism based on a thorough bifurcation analysis of the neuron's dynamics. The model incorporates HH--style equations to evolve the subthreshold currents and includes IF mechanisms to characterize spike events and mediate interactions between the subthreshold and spiking currents. The hybrid IF model successfully reproduces the dynamic behavior and temporal characteristics of the full model over a wide range of activity, including bursting and tonic firing. Comparisons of timed computer simulations of the reduced model and the original model for both single neurons and moderately sized networks (n < or = 500) show that this model offers improvement in computational speed over the HH--style bursting model.

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121. Lateral Neural Model of Binocular Rivalry

L Stollenwerk, Mathias Bode

https://doi.org/10.1162/089976603322518777 | pp. 2863-2882

This article introduces a two-dimensionally extended, neuron-based model for binocular rivalry. The basic block of the model is a certain type of astable multivibrator comprising excitatory and inhibitory neurons. Many of these blocks are laterally coupled on a medium range to provide a two-dimensional layer. Our model, like others, needs noise to reproduce typical stochastic oscillations. Due to its spatial extension, the noise has to be laterally correlated. When the contrast ratio of the pictures varies, their share of the perception time changes in a way that is known from comparable experimental data (Levelt, 1965; Mueller & Blake, 1989). This is a result of the lateral coupling and not a property of the single model block. The presentation of simple and suitable inhomogeneous stimuli leads to an easily describable perception of periodically moving pictures like propagating fronts or breathing spots. This suggests new experiments. Under certain conditions, a bifurcation from static to moving perceptions is predicted and may be checked and employed by future experiments. Recent "paradox" (Logothetis, 1999) observations of two different neuron classes in cortical areas MT (Logothetis & Schall, 1989) and V4 (Leopold & Logothetis, 1996), one that behaves alike under rivaling and nonrivaling conditions and another that drastically changes its behavior, are interpreted as being related to separate inhibitor neurons.

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122. Suprathreshold Intrinsic Dynamics of the Human Visual System

Gopathy Purushothaman, Haluk Ö■men, Harold E. Bedell

https://doi.org/10.1162/089976603322518786 | pp. 2883-2908

Intrinsic high-frequency neural activities have been observed in the visual system of several species, but their functional significance for visual perception remains a fundamental puzzle in cognitive neuroscience. Spatiotemporal integration in the human visual system acts as a low-pass filter and makes the psychophysical observation of high-frequency activities very difficult. A computational model of retino-cortical dynamics (RECOD) is used to derive experimental paradigms that allow psychophysical studies of high-frequency neural activities. A reduced-parameter version of the model is used to quantitatively relate psychophysical data collected in two of these experimental paradigms. Statistical analysis shows that the model's account of the variance in the data is, in general, highly significant. We suggest that psychophysically measured oscillations reflect intrinsic neuronal oscillations observed in the visual cortex.

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123. Closed-Form Expressions of Some Stochastic Adapting Equations for Nonlinear Adaptive Activation Function Neurons

Simone Fiori

https://doi.org/10.1162/089976603322518795 | pp. 2909-2929

In recent work, we introduced nonlinear adaptive activation function (FAN) artificial neuron models, which learn their activation functions in an unsupervised way by information-theoretic adapting rules. We also applied networks of these neurons to some blind signal processing problems, such as independent component analysis and blind deconvolution. The aim of this letter is to study some fundamental aspects of FAN units' learning by investigating the properties of the associated learning differential equation systems.

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124. Selecting Informative Data for Developing Peptide-MHC Binding Predictors Using a Query by Committee Approach

J K Christensen, Kasper Lamberth, Morten Nielsen, Claus Lundegaard, Peder Worning, Sanne Lise Lauemøller, Søren Buus, Søren Brunak, Ole Lund

https://doi.org/10.1162/089976603322518803 | pp. 2931-2942

Strategies for selecting informative data points for training prediction algorithms are important, particularly when data points are difficult and costly to obtain. A Query by Committee (QBC) training strategy for selecting new data points uses the disagreement between a committee of different algorithms to suggest new data points, which most rationally complement existing data, that is, they are the most informative data points. In order to evaluate this QBC approach on a real-world problem, we compared strategies for selecting new data points. We trained neural network algorithms to obtain methods to predict the binding affinity of peptides binding to the MHC class I molecule, HLA-A2. We show that the QBC strategy leads to a higher performance than a baseline strategy where new data points are selected at random from a pool of available data. Most peptides bind HLA-A2 with a low affinity, and as expected using a strategy of selecting peptides that are predicted to have high binding affinities also lead to more accurate predictors than the base line strategy. The QBC value is shown to correlate with the measured binding affinity. This

demonstrates that the different predictors can easily learn if a peptide will fail to bind, but often conflict in predicting if a peptide binds. Using a carefully constructed computational setup, we demonstrate that selecting peptides with a high QBC performs better than low QBC peptides independently from binding affinity. When predictors are trained on a very limited set of data they cannot be expected to disagree in a meaningful way and we find a data limit below which the QBC strategy fails. Finally, it should be noted that data selection strategies similar to those used here might be of use in other settings in which generation of more data is a costly process.

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125. Index

(authors unknown)

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